Employment and Earnings Trajectories of HUD Program Participants

by

Kyle Raze
U.S. Census Bureau

Rachel M. Shattuck
U.S. Census Bureau

David Pritchard
U.S. Census Bureau

Thomas B. Foster
U.S. Census Bureau

Sonya R. Porter
U.S. Census Bureau

Denise Flanagan-Doyle
U.S. Census Bureau

Veronica Garrison
U.S. Department of Housing and Urban Development

Jacqueline Bachand
U.S. Department of Housing and Urban Development

Ethan Krohn
U.S. Census Bureau

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Abstract

Federal housing assistance programs, such as those run by the U.S. Department of Housing and Urban Development (HUD), have been shown to reduce rent burden and improve housing stability for program participants, which may in turn have downstream impacts on their labor market attachment and career trajectories. However, existing studies from individual cities or states provide mixed evidence on the association of housing assistance with labor market outcomes. By linking HUD administrative records to matched employee-employer earnings records from the Longitudinal Employer-Household Dynamics (LEHD) program, we document how the labor market trajectories of program participants change as they enter and exit federal housing assistance programs, examining outcomes over a 14-year window surrounding entry or exit. In our analysis of entry, we find that the employment rates and earnings of first-time HUD program participants begin to increase upon entering a HUD program, which represents a reversal of prior declining trends in these outcomes. Suggestive of a positive association, these increases in employment and earnings trends exceed those of low-income non-participants from the American Community Survey (ACS). In our analysis of exits, we find that program participants who eventually leave a HUD program have increasing pre-exit trends in employment and earnings that then flatten upon exiting. Comparing these negative changes in trend to the relatively stable trajectories of those who remain in HUD programs throughout the analysis suggests that exits are associated with diminished employment and earnings trajectories.

Keyword: Housing Choice Voucher, project-based rental assistance, labor supply, in-kind transfers, linked employer-employee data

JEL Classification: I38, J22

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1. Introduction

A well-established social sciences literature demonstrates that housing and neighborhood conditions shape individual wellbeing and life outcomes. For residents of low-income neighborhoods, poor housing quality, overcrowding, crime, disorder, environmental hazards, and limited access to public services undermine physical and mental health as well as economic prospects (Bonnefoy, 2007; Jacob et al., 2011; Diez Roux and Mair, 2016; Aratani et al., 2019). For low-income families, difficulty securing affordable and adequately maintained housing creates stress and diverts time and resources away from other activities, including employment (Heintze et al., 2006; Fertig and Reingold, 2007). In this study, we contribute new longitudinal evidence on the employment and earnings outcomes of working-age adults in low-income families as they enter and exit housing assistance programs.

Federal housing assistance funded by the U.S. Department of Housing and Urban Development (HUD) is the central federal program addressing the housing needs of individuals and families with very low incomes. Public Housing Agencies (PHAs) administer HUD assistance to households facing significant disadvantages. Compared with both the general population and other low-income renters, recipients have lower incomes and educational attainment (Ryan and Bauman, 2016; Bellemore et al., 2024), include higher percentages of single female-headed households (Han and Waldfogel, 2001; Chaudry, 2006) and include higher shares of older adults and individuals with disabilities (Kingsley, 2017; Blakeslee et al., 2003; Taylor, 2018; Bellemore et al., 2024). As a result, many HUD recipients face circumstances that constrain their ability to enter and sustain stable employment.

There are several mechanisms through which housing assistance could help support recipients' employment. For example, by reducing the share of income spent on rent, subsidies can free resources for work-related expenses such as transportation, childcare, or clothing (Harkness and Newman, 2006). Assistance may also help households overcome barriers to work tied to health problems or childcare needs, as some PHAs supplement subsidies with services aimed at supporting job skills and labor force attachment (Chaudry, 2006; Aratani, 2019; Santiago and Leroux, 2022). In addition, housing assistance can make it possible for households to move to neighborhoods with lower poverty rates, greater access to jobs, and stronger public services (Kling et al., 2007; Jacob et al., 2011; Lens, 2014). However, such moves could also result in disruption of social networks or difficulty adapting to new environments (Clampet-Lundquist and Massey, 2008; Aratani, 2019).

While these mechanisms suggest that housing assistance could play a role in shaping labor market outcomes, the existing literature provides mixed evidence on its relationship to employment and earnings. Some studies find reductions in employment or earnings among recipients (Fischer, 2000; Jacob and Ludwig, 2012; Olsen et al., 2005; Susin, 2005), others find no effect (Bania et al., 2003; Van Ryzin et al., 2003), and still others report modest positive associations, often through stability or participation in supplemental programs such as Family Self-Sufficiency (FSS; Harkness and Newman, 2006; Heintze et al., 2006; Olsen et al., 2005). This variation may reflect data and design limitations: many studies focus on small samples, single cities, or short time horizons, and few follow HUD recipients longitudinally or compare them to similarly situated renters nationwide (Fischer, 2000; Harkness and Newman, 2006; Carlson et al., 2012). As a result, we have an incomplete picture of how employment and earnings track before, during, and after receipt of housing assistance.

In this study, we link HUD administrative records from 2005 to 2022 with Longitudinal Employer-Household Dynamics (LEHD) data from 2000 to 2023. These linked data allow us to follow the employment and earnings trajectories of HUD recipients over time, spanning periods before, during, and after receipt of assistance. To provide context, we also draw on the American Community Survey (ACS) to construct a comparison group of low-income renters not receiving HUD support. With this design, we document how recipients' labor market outcomes evolve as they enter assistance and compare the experiences of households that exit HUD programs with those that remain. Together, this approach provides new national, longitudinal evidence on the relationship between housing assistance, employment, and earnings, while also advancing methods for studying outcomes among disadvantaged populations.

Among first-time HUD entrants who enter a program between 2005 and 2014, we find that employment and earnings begin to increase after entry, reversing a pre-entry trend of decreasing employment and earnings. For example, in the four years before entry, entrants' earnings decline by an average of $112 per year. Conversely, in the ten years after entry, their earnings increase by an average of $174 per year. The difference in these trends indicates that slope of entrants' earnings trajectories increases by $286 per year, on average, after they enter a HUD program. Moreover, this post-entry trend break is $228 per year greater than that of low-income renters from an adjusted ACS comparison group, which suggests that the post-entry increase in the slope of entrant's earnings trajectories is not simply an artifact of broader changes in labor market conditions. Similarly, the slope of entrants' employment trajectories increases by an average of 1.45 percentage points per year after entry, a change that is 2.33 percentage points per year higher than that of the adjusted ACS comparison group. Although baseline employment and earnings levels differ substantially across sociodemographic groups, similar changes in labor market trajectories are evident within most groups of entrants, and these changes are typically large enough to reverse declining pre-entry trajectories.

For those who exit a HUD program between 2005 and 2011, employment and earnings generally rise, on average, both in the quarters leading up to exit and immediately afterward. Upon exiting, however, their employment and earnings trajectories begin to flatten. Differences between pre-exit and post-exit trends indicate that the slope of leavers' employment trajectories exit decreases by 1.05 percentage points per year and the slope of their earnings trajectories decreases by $244 per-year, on average. These trend breaks are 2.06 percentage points and $268 per year lower than those of an adjusted comparison group of individuals who remain in HUD programs throughout the follow-up period. As with entrants, employment and earnings levels differ across sociodemographic groups, but diminishing post-exit trajectories are evident within most groups of leavers.

While pre-existing differences in employment and earnings trends make it difficult to credibly predict how entrants and leavers would have fared had they not entered or exited a HUD program, post-entry and post-exit changes in employment and earnings trends—namely, the reversal of entrants' declining employment and earnings trajectories and the flattening of leavers' upward trajectories—provide suggestive evidence of a positive association between HUD program participation and participants' labor market trajectories.

We proceed in Section 2 with a discussion of HUD programs and the existing literature on housing assistance. Then, in Section 3, we describe the data sources we use to construct our analytical samples, which we outline in Section 4. After specifying our research methodology in Section 5, we then present results in Section 6 and conclude in Section 7.

2. Background on HUD Housing Assistance Programs and Recipients

2.A. Eligibility for and Receipt of Housing Assistance

HUD housing assistance is administered locally by Public Housing Authorities (PHAs) and generally serves households with very low incomes, defined as below 50 percent of Area Median Income (AMI). In some cases, households with incomes up to 80 percent of AMI may also qualify under special circumstances (Bellemore et al., 2024). Housing assistance is not an entitlement; only about 23% of eligible households receive subsidies (Kingsley, 2017). Applicants are typically placed on long wait lists, and PHAs prioritize those with the greatest need, though practices vary across agencies (Ambrose, 2005; Kingsley, 2017; Aratani et al., 2019).

HUD offers two primary forms of assistance. Tenant-based subsidies, most notably Housing Choice Vouchers (HCVs), account for about half of all assistance and subsidize private market units of a household's choice. Project-based assistance, where subsidies are tied to specific units, includes public housing managed by PHAs (about 20% of HUD subsidies) and project-based rental assistance through contracts with private owners. Assisted multifamily housing, where tenants rent units in properties contracted with PHAs, accounts for about 30% of HUD subsidies (Bellemore et al., 2024). Across both forms of assistance, tenants typically contribute 30 percent of their income toward rent, with subsidies covering the remainder up to a Fair Market Rent (FMR) standard. Households may pay the difference for higher-cost units (Bellemore et al., 2024).

2.B. Characteristics of the HUD-Assisted Population

While prior studies have not produced significance-tested national comparisons, our results and existing evidence show that HUD recipients differ substantially from both the general population and other low-income renters. They have lower educational attainment, fewer college graduates and more without a high school diploma (Ryan and Bauman, 2016; Bellemore et al., 2024) and lower household incomes than similarly poor but non-assisted renters (Bellemore et al., 2024). Single female-headed households are overrepresented, creating added childcare constraints (Han and Waldfogel, 2001; Chaudry, 2006). HUD households also include larger shares of older adults and individuals with disabilities with more than 30% of household heads being non-elderly individuals with disabilities. At the same time, older HUD residents report poorer health than older adults more broadly (Kingsley, 2017; Blakeslee et al., 2003; Bellemore et al., 2024; Taylor, 2018; Kler et al., 2024). Taken together, these characteristics are not only markers of disadvantage but also factors that could restrict labor market participation, creating structural barriers to employment among HUD recipients.

2.C. Housing and Neighborhood Changes Resulting from HUD Assistance

Beyond differences in recipient characteristics, housing assistance also shapes the environments in which households live. Assistance reduces rent burdens and provides greater residential stability, lowering risks of eviction and overcrowding (Brisson and Covert, 2015; Lundberg et al., 2021; Preston and Reina, 2021; Cai, 2024; Leung et al., 2023). Subsidies can improve housing quality through inspection standards and, in some cases, enable moves to lower-poverty neighborhoods, though such moves remain relatively rare (Abt Associates et al., 2006; Fertig et al., 2007; Kling et al., 2007; Carlson et al., 2012; Lens, 2014; Aratani et al., 2019).

PHAs and housing projects may also connect residents with social services, either directly or through case management (Fertig et al., 2007; Brisson and Covert, 2015). Despite these benefits, assisted housing is often located in disadvantaged neighborhoods with poorer maintenance and fewer economic opportunities (Kingsley, 2017; Fertig et al., 2017; Gabbe and Pierce, 2020). Moves into subsidized housing may also disrupt existing social networks (Aratani et al., 2019), and for some households the result may be environments that constrain rather than enhance employment prospects.

2.D. Lengths of Stay and Reasons for Exits

Another important dimension of HUD assistance includes how long households remain in programs and the circumstances under which they exit. Nationally, average stays are four to six years, though in some cities one-quarter of households remain for 20 years or more (Bahchieva and Hozier, 2001; McClure, 2018; Greenlee and McClure, 2024). Longer stays are more common among elderly residents, individuals with disabilities, female-headed households, and families with young or disabled children; shorter stays occur among higher-income or more educated households (Ambrose, 2005; Cortes et al., 2008; McClure, 2018). About 20 percent of households transition between program types, most often into HCVs (Greenlee and McClure, 2024).

Exits are typically classified as positive (increased income, homeownership), negative (eviction, lease violation), or neutral (moving in with relatives) (Smith et al., 2014; Matheson et al., 2022). Outcomes differ accordingly: negative exits are linked to worse health, food insecurity, and medical debt, while positive exits predict higher wages and greater neighborhood satisfaction (Smith et al., 2014; Matheson et al., 2022). Importantly, the same conditions influencing exit type such as disability, health, or rising wages may also shape post-exit outcomes.

2.E. Relationship of Housing Assistance to Employment, and Earnings

Potential Income and Employment Effects of Housing Assistance

A central question is how housing assistance shapes recipients' employment and earnings. On the one hand, assistance may support work by providing stability, reducing rent burdens, and freeing resources for employment-related expenses such as transportation and childcare (Harkness and Newman, 2006; Heintze et al., 2006). Vouchers may also facilitate moves to neighborhoods with better access to jobs, childcare, and transit (Bania et al., 2003; Blumenberg et al., 2015; Aratani et al., 2019). Parents may feel safer reducing child supervision and entering the labor force in lower-poverty neighborhoods (Susin, 2005). In addition, PHAs and programs like Family Self-Sufficiency (FSS) can connect households to training and allow them to retain increased earnings (Olsen et al., 2005; Santiago and Leroux, 2022).

On the other hand, housing assistance may discourage employment. Because eligibility requires income below program thresholds, recipients may face disincentives to increase earnings (Fischer, 2000; Smith et al., 2014). Rent subsidies can reduce financial pressure to work, and some assisted moves place families in neighborhoods with safety concerns or fewer job opportunities (Fischer, 2000; Susin, 2005; Heintze et al., 2006). Given that HUD recipients are disproportionately single mothers and individuals with disabilities, even stable housing may not

fully help individuals overcome barriers to labor force participation (Cortes et al., 2008; Matheson et al., 2022).

Evidence to Date on Housing Assistance, Employment, and Earnings

Research to date shows limited and mixed associations of housing assistance overall, and by type of assistance, with employment and earnings (Carlson et al., 2012). Some studies find reductions in employment or earnings among recipients (Fischer, 2000; Olsen et al., 2005; Susin, 2005; Jacob and Ludwig, 2012; Carlson et al., 2012). Others find no effect (Bania et al., 2003; Van Ryzin et al., 2003) or modest positive effects, particularly through stability or participation in FSS (Harkness and Newman, 2006; Heintze et al., 2006; Olsen et al., 2005). Results also vary by program type, subgroup, and time frame: tenant-based assistance often shows smaller negative effects than project-based housing (Olsen et al., 2005), and "positive" exits are linked with higher wages and more hours worked (Colombara et al., 2024).

Measurement Complications and Potential Mediating Factors

Differences across studies partly reflect data and methodological limitations. Studies vary in data sources (administrative vs. survey), geographies (local vs. national), and time periods. Many earlier studies focused on welfare leavers after the 1996 welfare reform, making generalization difficult (Fischer, 2000; Bania et al., 2003; Harkness and Newman, 2006). Few distinguish clearly between program types, despite evidence of heterogeneity (Olsen et al., 2005; Heintze et al., 2006; Carlson et al., 2012). Context also matters: access to jobs, transportation, and childcare varies widely across local housing markets and shapes employment opportunities (Joassart-Marcelli, 2005; Lens, 2014; Blumenberg et al., 2015; Chaudry, 2006; Ruppanner et al., 2019).

Below, we describe how we contribute to literature on HUD recipients' employment and earnings, both substantively and methodologically. Using HUD administrative records, we track entry and exit timing, and link to LEHD data to observe employment and earnings outcomes over up to 14 years. We also develop novel comparison groups, drawing on the ACS to identify low-income renters not receiving HUD assistance for the analysis of entry and HUD administrative records to identify program participants who remain in HUD programs for the analysis of exits.

3. Data

We examine the labor market trajectories of HUD program participants by linking person-level records across multiple administrative and survey datasets maintained by the U.S. Census Bureau. These include the HUD Public and Indian Housing Information Center/Tenant Rental Assistance Certification System (PIC/TRACS), the American Community Survey (ACS), the Longitudinal Employer-Household Dynamics (LEHD) program, the Master Address File Extract (MAF-X), the Auxiliary Reference File (MAF-ARF), the Census Numerical Identification File (Numident), and a collection of individual race and ethnicity responses drawn from the Decennial Census and ACS. These linkages are enabled by Protected Identification Keys (PIKs), anonymized unique identifiers developed by the Census Bureau to facilitate person-level integration across datasets while preserving confidentiality (Wagner and Layne, 2014).

Longitudinal HUD Records

We identify HUD program participants using the PIC/TRACS database, which tracks household participation in four types of rental assistance programs: Public Housing, Housing Choice Vouchers, Project-based Section 8, and other multifamily programs. These administrative data consist of transactions submitted by public housing agencies and private property managers as households enter, recertify, or exit HUD programs. Each transaction includes an action code indicating the type of transaction, along with detailed information on household income, demographics, disability status, and household composition. HUD standardizes these transactions into a longitudinal dataset released quarterly, with retrospective updates to account for late submissions. We use records from 2005 through 2022, covering all U.S. states and the District of Columbia. Nearly 99% of observations include a valid PIK and can be linked to other data sources.[1]

American Community Survey

We construct a comparison group of eligible non-participants from the American Community Survey (ACS), a nationally representative survey of more than 3.5 million households annually. We restrict to renters surveyed between 2005 and 2022 with incomes below 80 percent of Area Median Income (AMI), adjusted for household size and location. We adopt this broader cutoff, rather than HUD's stricter thresholds of 50 or 30 percent of AMI, to limit regression-to-the-mean bias, since ACS measures income at a single point in time and narrower cutoffs would disproportionately capture households experiencing temporary setbacks from which they eventually recover (Daw and Hatfield 2018).[2] Among respondents meeting these criteria, only 84.7 percent have valid PIKs, and because PIK-linked individuals differ from the full ACS sample (Bond et al. 2014), we reweight them to restore national representativeness (Raze et al., 2026).[3,4] We then exclude anyone appearing in HUD records, yielding a comparison group of eligible non-participants.

Longitudinal Employer-Household Dynamics

To measure labor market outcomes, we link HUD participants and ACS comparators to the Longitudinal Employer-Household Dynamics (LEHD) program, a linked employee–employer database derived from state unemployment insurance records that covers more than 95 percent of civilian wage and salary jobs in the United States, excluding only groups such as federal employees, the self-employed, independent contractors, and some family workers (Graham et al., 2022).[5] We use LEHD data from 2000 to 2023 to construct quarterly employment

[1] In Appendix Table A.1, we show high PIK rates across age groups, sexes, races, disability, regions, and program types.
[2] In Appendix E, we report alternative estimates using very low-income and extremely low-income ACS comparison groups.
[3] In Appendix Table A.2, we show PIK rate varies by age, sex, race, and disability status.
[4] We apply Inverse Probability Tilting (IPT) to weight PIK-linked low-income renters so they match the socioeconomic profile of the full low-income renter population (Graham et al. 2012). Specifically, we reweight linkable low-income renters to match the full set of low-income renters within each ACS response quarter on sex, race, disability status, citizenship, household income, and a 4th-order polynomial in age.
[5] As we show in Appendix B, LEHD earnings tend to align well, on average, with the individual wage earnings collected by PHAs during the certification and recertification process.

and earnings histories for all individuals with valid Protected Identification Keys (PIKs). Following standard practice (e.g., Chyn et al., 2025), employment is defined as having positive UI-covered earnings in a given quarter, and all earnings are expressed in 2022 dollars using the R-CPI-U-RS (Stewart and Reed, 1999).[6] To avoid bias from conditioning on current employment, we include zeroes for non-working quarters. Among those with valid PIKs, the vast majority of working-age HUD participants and ACS comparators appear in the LEHD at least once during the observation window.[7]

Other Sources

To assign geographic identifiers to HUD program participants, we combine information from the Master Address File Extract (MAF-X), the Auxiliary Reference File (MAF-ARF), and publicly available geocoded HUD project addresses.[8] We link the Master Address File Identifier (MAFID, the Census Bureau's anonymized address code) to the MAF-X to assign state identifiers.[9] We use the Census Numident to identify deaths and restrict the sample to individuals who were alive for at least ten years after entering or exiting HUD. We do this to exclude cases where early mortality could bias results upward, since individuals who die sooner are likely to have lower employment or earnings. The Numident also supplements missing values for date of birth and sex, which are sometimes unavailable in HUD records. Race and ethnicity are drawn from a linked file of individual responses to Decennial Census and ACS questionnaires, and for individuals who do not link to either auxiliary source we use their most recent valid HUD reported values.

4. Sample Construction

From the linked data, we construct two longitudinal samples: one that follows individuals before and after their first entry into HUD assistance (entry sample), and another that follows them before and after their final exit (exit sample). These samples allow us to observe labor market outcomes around program transitions while holding the population of interest fixed. We assign date of entry or exit using HUD records, with imputations where dates are missing.[10] For each individual, we build a balanced panel of 56 quarters, spanning 16 quarters before and 40 quarters after the quarter of entry or exit. The analysis is limited to working-age

[6] Specifically, we use the "All Items" series available at https://www.bls.gov/cpi/research-series/r-cpi-u-rs-home.htm.

[7] We provide LEHD match rates for each of the analysis samples we outline further below in Appendix Table C.1 and Appendix Table C.2. Apart from individuals with disabilities and some groups of individuals over the age of 44, match rates tend to exceed 90 percent. A failure to match indicates that an individual did not participate in covered employment between 2000 and 2023.

[8] We obtain data on housing project addresses from HUD's Picture of Subsidized Households database, which is available at https://www.huduser.gov/portal/datasets/assthsg.html.

[9] For households without a valid MAFID, we follow a hierarchy of alternatives, beginning with geocoded unit coordinates in the HUD data, followed by geocoded project addresses matched conservatively to HUD records, state identifiers derived from contract, development, or PHA codes, and finally MAFIDs from the MAF-ARF linked through head of household PIKs.

[10] For the entry sample, we define the entry date as the earliest valid admission date or, if missing, the effective date of the first HUD transaction. For the exit sample, the exit date is based on the last HUD transaction with an exit code; when this is missing, we impute the exit date as 12 months after the final effective date, or 36 months for participants in Moving to Work agencies, which have longer recertification intervals.

adults who are between ages 18 and 64 throughout the window, who survive through the end of the observation period, and who have at least one appearance in the LEHD.

Entry Analysis Sample

We define an "entry cohort" as the calendar quarter in which a HUD participant first appears in program records or an ACS respondent completes the survey.[11] We then track labor market outcomes relative to that quarter, yielding 37 distinct cohorts. Because HUD participants are not evenly distributed across cohorts, ACS respondents must be reweighted to avoid overrepresenting periods with relatively few HUD entrants.[12] Following Wing et al. (2024), we adjust ACS person weights so that the timing distribution of comparison groups aligns with that of HUD participants.[13]

Our entry sample includes 2,086,000 individuals who entered HUD between the first quarter of 2005 and the first quarter of 2014, as well as up to 1,776,000 eligible non-participants from the ACS, representing 1,747,000 unique individuals who responded to the survey during the same period.[14] Combined, these observations span 216,272,000 person-cohort-quarter-year units. We define *HUD entrants* as individuals appearing in HUD records for the first time during this period, and *low-income ACS comparators* as renters in the ACS with household incomes below 80 percent of Area Median Income who do not appear in HUD administrative data. Table 1 (columns 1, 2, and 4) shows that while both groups are economically disadvantaged, HUD entrants are notably worse off prior to receiving housing assistance. In the 16 quarters before entry, entrants have an average employment rate of 50%, which is 11 percentage points lower than that of comparators. Their average quarterly earnings over the same period are $2,620, more than $2,300 below those of comparators. At the time of entry, HUD entrants report household incomes that are $18,500 lower on average, partly reflecting smaller household sizes. Demographic differences reinforce this pattern of disadvantage. HUD entrants are significantly more likely than comparators to report a disability, to be female, and to identify as non-Hispanic Black or non-Hispanic American Indian or Alaska Native. These characteristics are strongly correlated with labor market disadvantage, suggesting that HUD entrants face deeper barriers to stable employment even before receiving assistance.

Given the systematic baseline differences between HUD entrants and low-income ACS comparators, we reweight the ACS sample to create a more comparable group. Using Inverse Probability Tilting (IPT), we adjust weights within each entry cohort so that comparators align

[11] Most ACS non-participants contribute a single response, but those observed in multiple survey years may enter more than one cohort.

[12] Without reweighting, ACS cohorts with larger samples but fewer HUD entrants would disproportionately influence the comparison, producing biased estimates.

[13] Specifically, we multiply the ACS person weights of eligible non-participants by $\frac{N_c^{HUD}/N^{HUD}}{N_c^{ACS}/N^{ACS}}$, where N_c^{HUD} is the number of program participants in entry cohort c, N^{HUD} is the total number of program participants in the analysis sample, N_c^{ACS} is the weighted number of eligible non-participants in entry cohort c, and N^{ACS} is the total weighted number of eligible non-participants in the analysis sample.

[14] All sample sizes and estimates in this report have been rounded to comply with Census Bureau disclosure avoidance guidelines.

with entrants on demographic and household characteristics.[15] As shown in Table 1 (columns 3 and 5), the IPT procedure reduces baseline imbalances across observable characteristics. Although level differences in employment and earnings remain, Section IV demonstrates that adjusted comparators display pre-entry trends more like those of HUD entrants, improving the validity of the comparison.[16]

Exit Analysis Sample

We define an exit cohort as the calendar quarter in which a leaver has their final HUD record. For each cohort, we identify stayers who entered at least 16 quarters before the leaver's exit and who have no record of leaving during the 40 quarters after the leaver's exit.[17] Stayers may appear in more than one cohort if they meet the criteria multiple times, but leavers only appear once. In total, we examine 25 exit cohorts, and, as in the entry analysis, we reweight ACS comparators using the method of Wing et al. (2024) so that the distribution of stayers aligns with that of leavers.[18, 19]

Our exit analysis includes 1,233,000 leavers who left a HUD program for the last time between the first quarter of 2005 and the first quarter of 2011, and 29,750,000 stayers (representing 1,882,000 unique individuals) who remained in HUD throughout the analysis window, yielding a total of 1,735,048,000 person–cohort–quarter–year observations. Table 2 (columns 1, 2, and 4) shows that leavers exhibit stronger baseline labor force attachment, with an average employment rate over 59%, nearly 13 percentage points higher than stayers, and average quarterly earnings of $3,745, exceeding those of stayers by $1,354. Demographic profiles also differ, leavers tend to be younger, are more likely to be male, and include a larger share of non-Hispanic White and a smaller share of non-Hispanic Black individuals. Program participation varies as well, with a lower percentage of leavers coming from public housing and a higher percentage from vouchers or other programs. By the time of exit, leavers had also spent relatively less time in HUD assistance than stayers.

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[15] Specifically, we use IPT (Graham et al. 2012; Sant'Anna and Zhao 2020) to enforce exact balance on sex, race, disability status, household size by age group (0–4, 5–12, 13–17, 18+), census region, and a fourth-order polynomial in age.

[16] We cannot reweight on several characteristics such as education, occupation, transportation, or detailed measures of disability and family structure because they are not available or are difficult to measure in HUD data. Appendix D shows that working-age adults who enter HUD differ significantly from those who never enter on these dimensions.

[17] To avoid including stayers with imputed or mismeasured exit dates, we further require that they show no evidence of leaving in the 12 quarters following the analysis window. This 12-quarter restriction is based on the 36-month imputation rule for program participants in Moving to Work PHAs.

[18] Similar to our approach in constructing the entry analysis sample, we assign each stayer in exit cohort c a sample weight equal to $\frac{N_c^{Leaver}/N^{Leaver}}{N_c^{Stayer}/N^{Stayer}}$, where N_c^{Leaver} is the number of leavers in exit cohort c, N^{Leaver} is the total number of leavers in the analysis sample, N_c^{Stayer} is the number of stayers in exit cohort c, and N^{Stayer} is the total number of stayers in the analysis sample.

[19] Again, we reweight stayers using IPT techniques. Our adjustment uses the same covariates as in the entry analysis, including sex, race, disability status, household size by age group, census region, and a fourth-order polynomial in age, and incorporates program type and the time between entry and exit. As reported in Table 2, columns 3 and 5, the IPT procedure reduces baseline imbalances in observable characteristics between leavers and stayers. Although some level differences remain, Section IV shows that adjusted stayers exhibit pre-exit employment and earnings trends that more closely track those of leavers.

Table 1: Baseline characteristics of HUD entrants and low-income ACS comparators

	HUD entrants	Low-income ACS comparators		Difference (HUD - ACS)	
	Mean	Mean	Adjusted mean	Unadjusted	Adjusted
Baseline outcomes					
Employment rate	50.06%	60.05%	61.48%	-9.99 p.p. *	-11.42 p.p. *
Quarterly wage earnings	$2,620	$4,867	$4,860	-$2,247 *	-$2,240 *
Disability status					
Does not have a disability	80.71%	90.09%	80.71%	-9.38 p.p. *	0.00 p.p.
Has a disability	19.29%	9.91%	19.29%	9.38 p.p. *	0.00 p.p.
Sex					
Female	65.76%	49.53%	65.76%	16.23 p.p. *	0.00 p.p.
Male	34.24%	50.47%	34.24%	-16.23 p.p. *	0.00 p.p.
Race					
Non-Hispanic white	43.04%	48.54%	43.04%	-5.49 p.p. *	0.00 p.p.
Non-Hispanic Black	34.22%	14.80%	34.22%	19.42 p.p. *	0.00 p.p.
Hispanic of any race	16.10%	27.77%	16.10%	-11.67 p.p. *	0.00 p.p.
Non-Hispanic Asian	2.49%	5.95%	2.49%	-3.45 p.p. *	0.00 p.p.
Non-Hispanic two or more races	2.28%	1.59%	2.28%	0.68 p.p. *	0.00 p.p.
Non-Hispanic AIAN	1.29%	0.78%	1.29%	0.51 p.p. *	0.00 p.p.
Non-Hispanic other race	0.57%	0.58%	0.57%	0.00 p.p.	0.00 p.p.
Age					
Age at entry or interview (years)	34.75	34.58	34.75	0.17 *	-0.00
Census region					
South	37.30%	34.23%	37.30%	3.07 p.p. *	0.00 p.p.
Midwest	24.82%	19.26%	24.82%	5.57 p.p. *	0.00 p.p.
Northeast	19.03%	16.98%	19.03%	2.05 p.p. *	0.00 p.p.
West	18.85%	29.53%	18.85%	-10.68 p.p. *	0.00 p.p.
Household characteristics					
Household income	$19,160	$38,010	$32,080	-$18,850 *	-$12,920 *
Number of adults	1.45	1.99	1.45	-0.54 *	0.00
Number of young children (0-4)	0.53	0.36	0.53	0.17 *	-0.00
Number of children (5-12)	0.58	0.46	0.58	0.11 *	-0.00
Number of older children (13-17)	0.22	0.23	0.22	-0.00 *	0.00
HUD program					
Housing Choice Vouchers	46.81%				
Public housing	27.22%				
Project-based Section 8	22.72%				
Other multifamily	3.26%				
Sample sizes					
Observations	2,086,000	1,773,000	1,773,000	3,860,000	3,860,000
Unique PIKs	2,086,000	1,745,000	1,745,000	3,831,000	3,831,000

Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Baseline outcomes are measured during the 16 quarters leading up to the quarter of entry or interview. All dollar amounts are adjusted to 2022 levels using the R-CPI-U-RS. Adjusted statistics are obtained by reweighting comparators to match the mean non-outcome characteristics of HUD entrants within each quarterly entry cohort. Asterisks (*) indicate that an estimated difference is statistically significant at the 5-percent level, based on a statistical test that accounts for heteroskedasticity and clustering at the PIK level. AIAN = American Indian/Alaska Native.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Table 2: Baseline characteristics of HUD leavers and stayers

	HUD leavers	HUD stayers		Difference (leavers - stayers)	
	Mean	Mean	Adjusted mean	Unadjusted	Adjusted
Baseline outcomes					
Employment rate	59.50%	46.79%	48.18%	12.71 p.p. *	11.31 p.p. *
Quarterly wage earnings	$3,745	$2,391	$2,314	$1,354 *	$1,431 *
Disability status					
Does not have a disability	93.78%	87.04%	93.78%	6.74 p.p. *	0.00 p.p.
Has a disability	6.22%	12.96%	6.22%	-6.74 p.p. *	0.00 p.p.
Sex					
Female	70.31%	80.59%	70.31%	-10.28 p.p. *	0.00 p.p.
Male	29.69%	19.41%	29.69%	10.28 p.p. *	0.00 p.p.
Ethnicity					
Non-Hispanic white	30.91%	23.77%	30.91%	7.13 p.p. *	0.00 p.p.
Non-Hispanic Black	44.40%	53.34%	44.40%	-8.93 p.p. *	0.00 p.p.
Hispanic of any race	17.92%	17.21%	17.92%	0.71 p.p. *	0.00 p.p.
Non-Hispanic Asian	2.90%	2.09%	2.90%	0.81 p.p. *	0.00 p.p.
Non-Hispanic two or more races	2.27%	2.27%	2.27%	0.00 p.p.	0.00 p.p.
Non-Hispanic AIAN	1.14%	0.89%	1.14%	0.24 p.p. *	0.00 p.p.
Non-Hispanic other race	0.46%	0.42%	0.46%	0.04 p.p. *	0.00 p.p.
Age and program duration					
Age at leavers' exit (years)	33.64	37.53	33.64	-3.89 *	0.00
Quarters in HUD at leavers' exit	42.40	46.46	42.40	-4.06 *	0.00
Census region					
South	37.15%	32.93%	37.15%	4.23 p.p. *	0.00 p.p.
Midwest	21.79%	21.59%	21.79%	0.20 p.p. *	0.00 p.p.
Northeast	21.64%	27.66%	21.64%	-6.02 p.p. *	0.00 p.p.
West	19.41%	17.82%	19.41%	1.59 p.p. *	0.00 p.p.
Household characteristics					
Number of adults	1.48	1.42	1.48	0.07 *	-0.00
Number of young children (0-4)	0.56	0.49	0.56	0.07 *	0.00
Number of children (5-12)	0.84	0.77	0.84	0.08 *	-0.00
Number of older children (13-17)	0.47	0.44	0.47	0.02 *	-0.00
HUD program					
Housing Choice Vouchers	49.81%	47.41%	49.81%	2.40 p.p. *	0.00 p.p.
Public housing	24.79%	28.54%	24.79%	-3.75 p.p. *	0.00 p.p.
Project-based Section 8	21.79%	21.10%	21.79%	0.69 p.p. *	0.00 p.p.
Other multifamily	3.50%	2.88%	3.50%	0.62 p.p. *	0.00 p.p.
Other	0.11%	0.08%	0.11%	0.03 p.p. *	0.00 p.p.
Sample sizes					
Observations	1,233,000	29,750,000	29,750,000	30,990,000	30,990,000
Unique PIKs	1,233,000	1,882,000	1,882,000	3,115,000	3,115,000

Notes: HUD leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. HUD stayers are working-age adults who remained in a HUD program throughout the analysis. Stayers can serve as comparators for multiple cohorts of leavers, and the inclusion criteria leads most to appear in the sample more than once (see Section 4 for details). Baseline outcomes are measured during the 16 quarters leading up to the leavers' quarter of exit. All dollar amounts are adjusted to 2022 levels using the R-CPI-U-RS. Adjusted statistics are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort. Asterisks (*) indicate that an estimated difference is statistically significant at the 5-percent level, based on a statistical test that accounts for heteroskedasticity and clustering at the PIK level. AIAN = American Indian/Alaska Native.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7 505232: CBDRB-FY26-028).

5. Methodology

We use the entry and exit analysis samples to characterize employment and earnings profiles of HUD participants before, during, and after program participation. The entry analysis compares first-time entrants with eligible non-participants in the ACS, while the exit analysis compares HUD leavers with stayers. For both comparisons, we estimate quarterly employment rates and mean earnings over the 16 quarters prior to entry or exit and across the 40 quarters thereafter.

Entry Analysis

To illustrate the average labor market trajectory of entrants before and after they enter a HUD program, we regress the employment or earnings Y_{ict} of entrant i in entry cohort c observed *t* quarters after entry on a series of quarter-relative-to-entry indicators $\sum_{\tau=-16}^{39} \mathbb{1}(\mathrm{t} = \tau)$, estimating models of the form

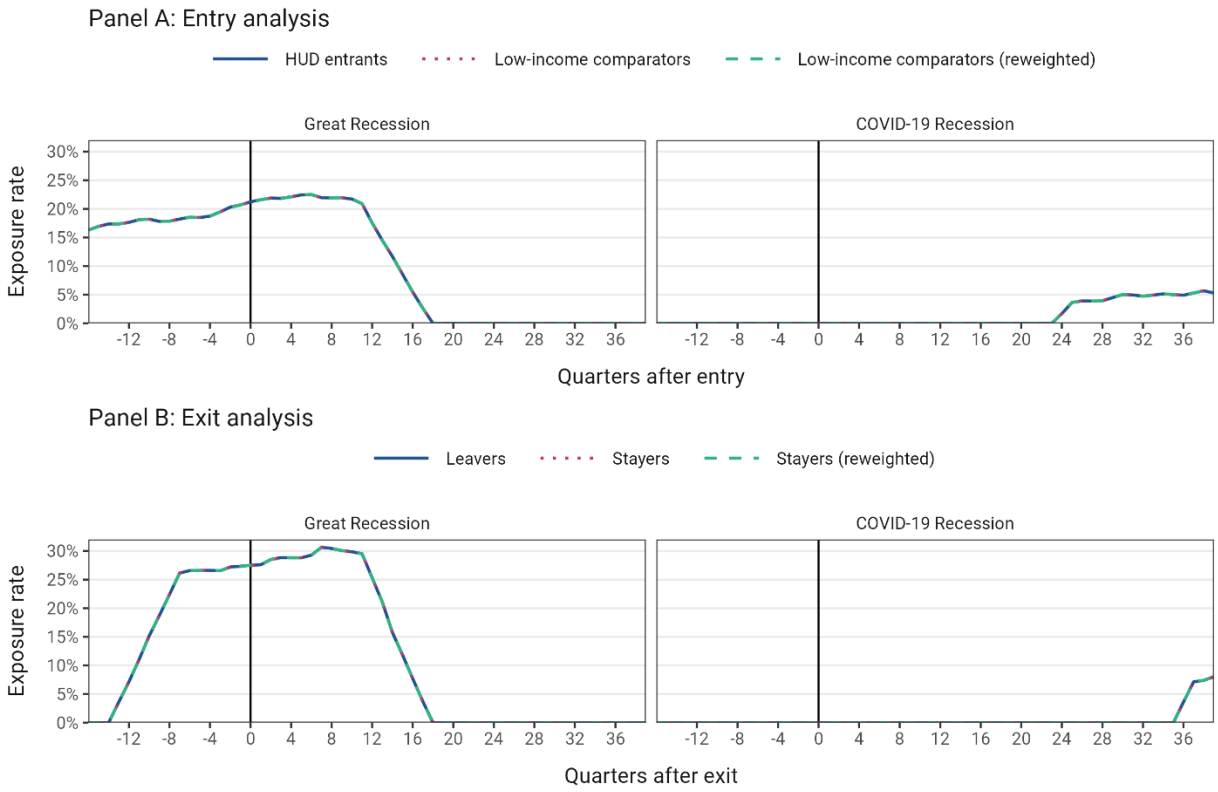
$$Y_{ict} = \sum_{\tau=-16}^{39} \alpha_\tau \mathbb{1}(\mathrm{t} = \tau) + \varepsilon_{ict} \,, \tag{1}$$

where the coefficients α_τ capture the employment rate or average earnings at each relative quarter and the error term ε_{ict} allows arbitrary correlation within cohorts over time. We estimate each model using ordinary least squares (OLS) and report cluster-robust 95% confidence intervals for each $\hat{\alpha}_\tau$ estimate. This regression-based approach allows us to flexibly trace labor market trajectories before and after entry while quantifying uncertainty in a way that accommodates heteroskedasticity and autocorrelation within cohorts.

As a precursor to estimating the association of entry into a HUD program with labor market trajectories, we also estimate a version of Equation (1) for low-income ACS comparators, reporting both unadjusted estimates and adjusted estimates that reweight the sample of comparators to match HUD entrants on disability status, sex, race, age, Census region, and household composition. Given that our sample period encompasses two major recessions, a simple before-and-after comparison of entrants' labor market outcomes alone could conflate the effects of entry with recession-induced changes in labor market conditions. Indeed, as we show in Panel A of Figure 1, the fraction of HUD entrants observed during either the Great Recession or the COVID-19 Recession varies over time relative to the quarter of entry, which could influence entrants' observed labor market trajectories. Comparing entrants to ACS comparators helps adjust for recession exposure, as the fraction of ACS comparators observed during either recession is identical to that of HUD entrants over relative time (as evidenced by the perfect overlap in recession exposure rates between entrants and low-income comparators in Panel A of Figure 1).

Even with an adjusted ACS comparison group that matches HUD entrants on sociodemographic characteristics and recession exposure, pre-existing differences in employment and earnings present challenges for credibly estimating the association of entry with labor market outcomes. For example, comparing the post-entry labor market outcomes of entrants with those of ACS comparators would conflate the effects of entry with pre-existing differences in employment and earnings, such as those documented in Table 1. A valid comparison must account for those pre-existing differences.

Figure 1: Exposure to macroeconomic events



Notes: Exposure rates represent the proportion of individuals who are observed during either the Great Recession (Q4 of 2007 through Q2 of 2009) or the COVID-19 Recession (Q1 and Q2 of 2020). In Panel A, HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1 and low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted statistics for low-income comparators are obtained by reweighting comparators to match the mean non-outcome characteristics of HUD entrants within each quarterly entry cohort. In Panel B, leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1 and stayers are working-age adults who remained in a HUD program throughout the analysis. Adjusted statistics for stayers are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Comparing before-and-after changes in labor market outcomes between entrants and ACS comparators—a difference-in-differences comparison—would control for pre-existing differences in the employment and earnings *levels*, provided that there are no pre-existing differences in employment and earnings *trends*. However, as will we show in Section VI, entrants and ACS comparators exhibit different employment and earnings trends before entry, which would lead a difference-in-differences comparison to conflate the effects of entry with pre-existing differences in employment and earnings trends. Credibly estimating the association of entry with labor market outcomes therefore requires a different approach.

We estimate the association of entry with labor market trajectories by comparing the change in entrants' employment and earnings trends after entry with that of ACS comparators. In contrast to difference-in-means and differences-in-differences comparisons, this comparison

adjusts for pre-existing differences in the levels and trends of each outcome. To make the comparison, we regress the employment or earnings Y_{ict} of individual i in entry cohort c observed t quarters after entry on an entrant indicator E_i (equal to one for entrants or zero for ACS comparators) interacted with a linear time trend t and a hinge function $t \times \mathbb{1}(\mathrm{t} \geq 0)$ that allows the slope of the time trend to change following the quarter of entry:

$$Y_{ict} = \underbrace{\beta_0 + \beta_1 E_i}_{\text{Baseline levels}} + \underbrace{\beta_2 t + \beta_3 t \times E_i}_{\text{Baseline trends}} + \underbrace{\beta_4 t \times \mathbb{1}(\mathrm{t} \geq 0) + \beta_5 t \times \mathbb{1}(\mathrm{t} \geq 0) \times E_i}_{\text{Post-entry trend breaks}} + \nu_{ict} \,. \quad (2)$$

In this interacted linear spline regression model, the intercept coefficients β_0 and β_1 capture pre-existing level differences in outcomes, the time trend coefficients β_2 and β_3 capture pre-existing differences in outcome trends, the trend break coefficient β_4 represents the change in the slope of the trend time for ACS comparators after the quarter of entry (e.g., due to changes in labor market conditions), and the "difference-in-trend-breaks" coefficient β_5 represents the differential change in the slope of time trend for entrants after the quarter of entry—that is, the association of entry with entrants' post-entry labor market trajectories. As in Equation 1, the error term ν_{ict} allows arbitrary correlation within cohorts over time. We estimate the model using Weighted Least Squares (WLS), where we weight each entrant by one and each comparator by their adjusted ACS person weight, and report cluster-robust *p*-values for each $\hat{\beta}_5$ estimate of the adjusted association.

To interpret the estimated associations as causal, one must assume that there are no remaining differences in unmeasured characteristics or circumstances between entrants and ACS comparators after adjusting for differences in sociodemographic characteristics and pre-existing differences in the levels and (linear) trends of employment and earnings. While this is a strong assumption, some institutional features of HUD programs help limit the scope for endogenous selection into assistance. For example, many housing authorities maintain long waitlists, and the timing of entry often depends more on unit turnover and administrative processes than abrupt changes in individual circumstances (Ambrose 2005; Aratani 2019). However, as we demonstrate in a supplemental analysis of the ACS in Appendix D, there are significant differences between eventual HUD participants and never-participating low-income renters in educational attainment, work history, family structure, and disability severity which we are unable to measure and adjust for in the full entry analysis sample. If the adjustments for baseline outcome levels and trends in Equation 2 fail to adequately adjust for these unmeasured differences, then the estimated association would fail to isolate the causal effect of entry. In any case, we view our difference-in-trend-breaks comparison as less biased than a simple difference-in-means comparison or a difference-in-differences comparison.

Exit Analysis

We perform a parallel analysis for leavers and stayers using versions of the specifications in Equation 1 and Equation 2 that are centered on the quarter of exit. Since both leavers and stayers participate in HUD programs at baseline, they share unobserved characteristics related to program entry and overlapping participation spells, which makes them more comparable than the never-participating low-income ACS renters used in the entry analysis. However, the exit analysis may be relatively more vulnerable to selection bias if the timing of exits reflects self-selection based on changes in individual circumstances. For example, some participants leave HUD when higher earnings make them ineligible, and these individuals may continue to have more favorable outcomes than those who remain (Matheson

et al., 2022). Furthermore, as we document in a companion report (Pritchard et al., forthcoming), we are unable to distinguish between positive, negative, and neutral exits for the vast majority of leavers in our data, so our exit analysis necessarily pools a variety of reasons for exiting a program.

Taken together, the entry and exit analyses provide a longitudinal perspective on the relationship between HUD program participation and labor market dynamics. By following the same individuals before and after entry or exit, our comparisons avoid biases from compositional changes in the HUD population. This longitudinal perspective also allows adjusting for pre-existing differences in labor market trajectories between comparison groups, reducing bias in estimates of the association of entry or exit with subsequent labor market trajectories.

6. Results

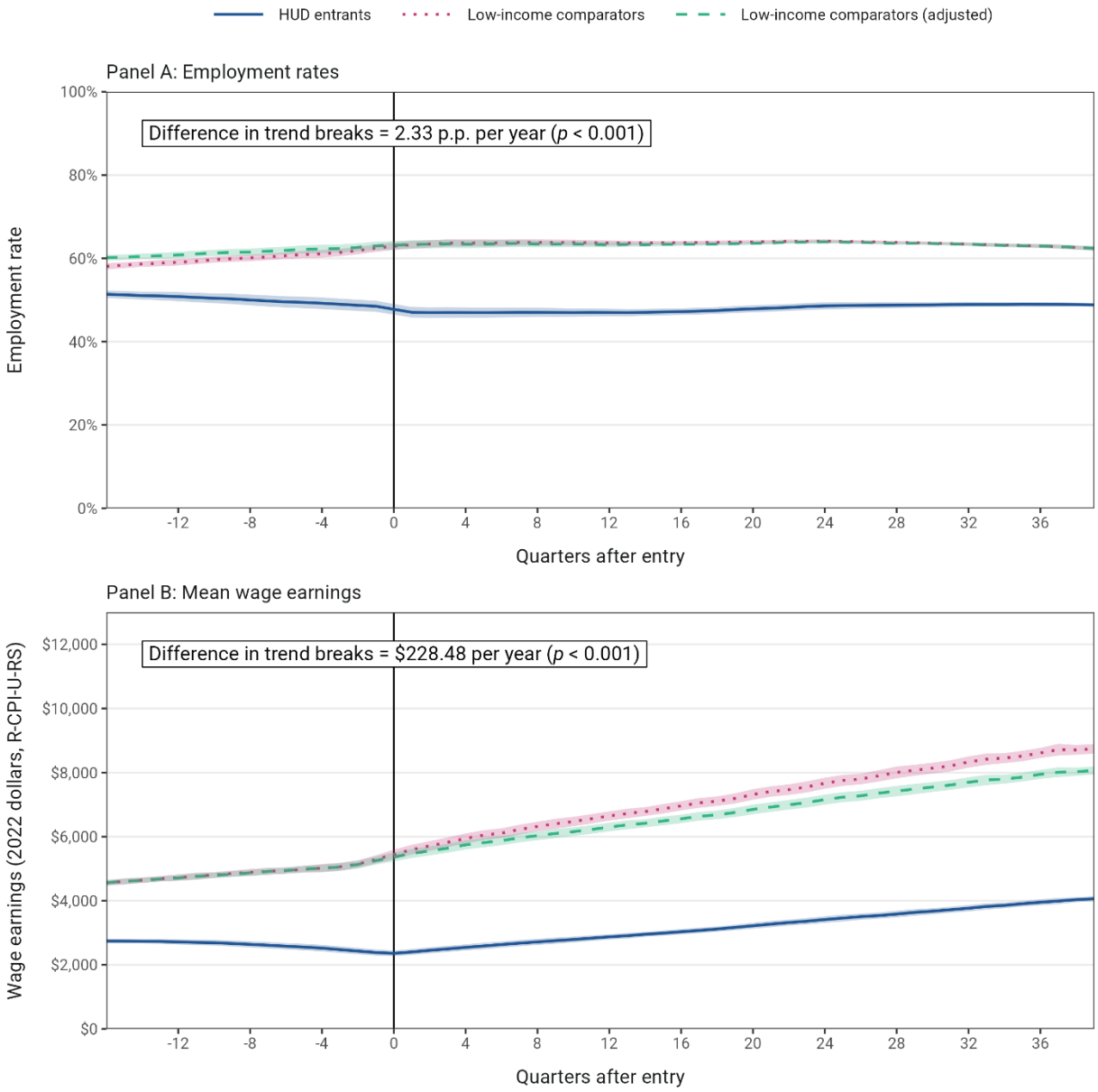
6.A. Entry Analysis

Full-Sample Trajectories

We present average labor market trajectories for HUD entrants and low-income ACS comparators in Figure 2. In Panel A we show that entrants and their low-income comparators have diverging trends in employment in the quarters leading up to the quarter of entry. Entrants have pre-entry employment rates that decrease by 1.25 percentage points per year, on average, whereas (adjusted) low-income comparators have employment rates that increase by an average of 0.87 percentage points per year during the same period. After entry, we find that the trend in employment for entrants continues downward for one quarter before recovering and trending slightly upward thereafter; over the entire 10-year follow-up period, entrants' employment rates increase by an average of 0.19 percentage points per year, which represents a 1.45 percentage point per year increase over their pre-entry employment trend. If coincident changes in macroeconomic conditions were driving the positive change in entrants' employment trajectories, then we would also expect low-income comparators to experience positive changes in their employment trajectories. However, entrants' post-entry change in their employment trend exceeds that of low-income comparators by 2.33 percentage points per year, which suggests that something other than broader changes in macroeconomic conditions explains the positive change in entrants' employment trajectories.

In Panel B we show that earnings exhibit a similar pattern. As with employment, we find diverging trends in earnings in the quarters leading up to the quarter of entry. Entrants' quarterly earnings decrease at an annualized rate of $112 per year, on average, whereas the earnings of (adjusted) low-income comparators increase by an average of $230 per year. After entry, we find that entrants' earnings increase by an average of $174 per year, which represents a $286 per year increase over their pre-entry earnings trend. This change in trend is $228 per year greater than that for low-income comparators, which suggests that entry is associated with positive changes in entrants' earnings trajectories.

Figure 2: Labor market trajectories of HUD entrants and low-income ACS comparators



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

While entrants' employment and earnings exhibit a nadir around the time they first enter a HUD program—likely a time when households are experiencing particularly difficult circumstances, which for some may include serious health crises or homelessness (Brisson and Covert, 2015)—these results indicate that entry is associated with improved post-entry labor market trajectories among entrants. The differential increase in employment trajectories after entry suggests that some combination of greater housing stability (Lundberg et al., 2021), improved access to services (Olsen et al., 2005; Fertig et al., 2007; Santiago and Leroux, 2022), freed household resources (Fisher, 2000; Susin, 2005), or other downstream effects of relocation (Kling et al., 2007; Carlson et al., 2012) may facilitate stronger labor force attachment among entrants. Similarly, the upward pivot in earnings trajectories suggests that housing assistance may also enable increases in hours worked or transitions into better-paying jobs.

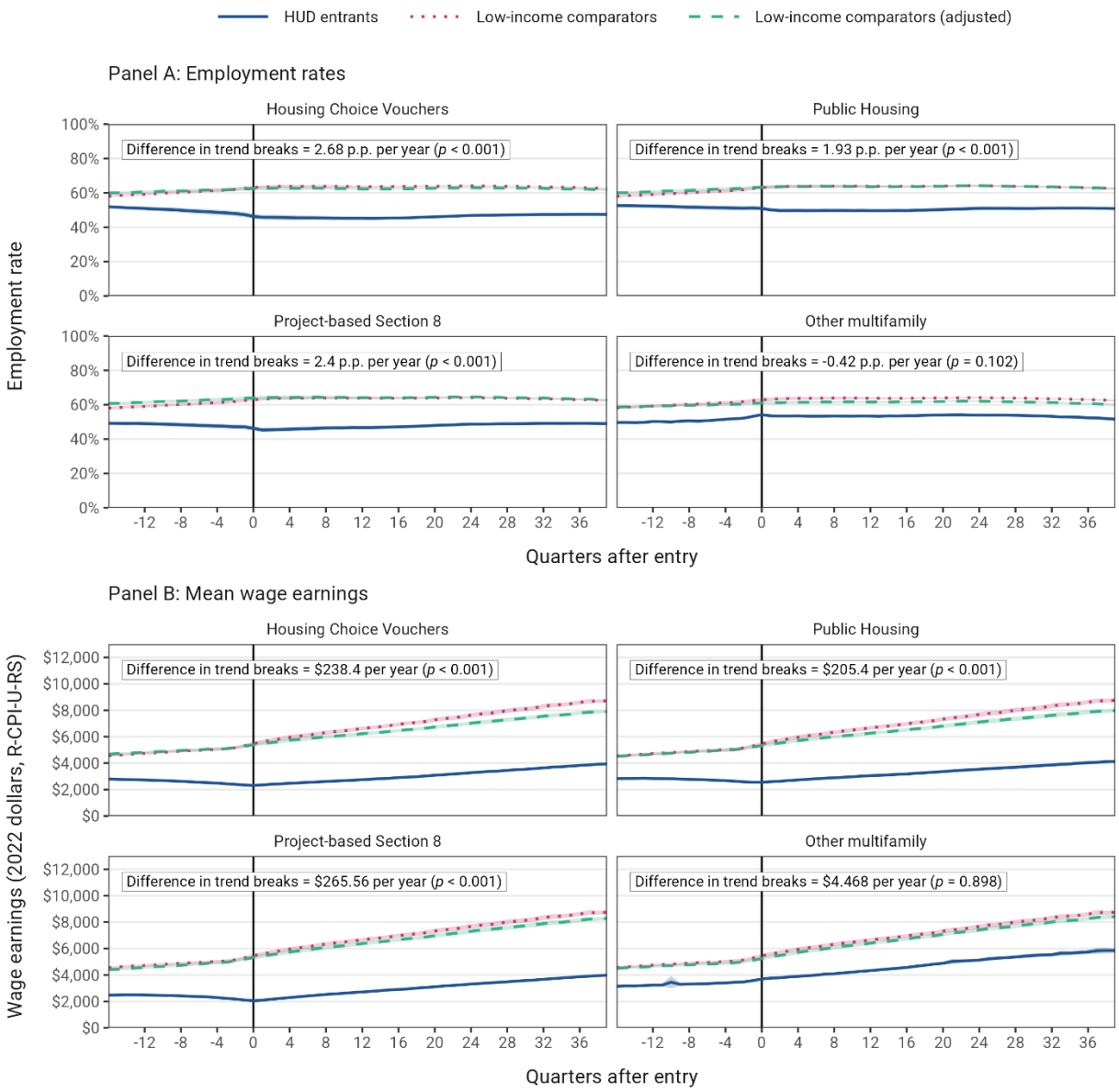
Group-Specific Trajectories

In Figures 3–10, we present the labor market trajectories of entrants and low-income comparators disaggregated by program type, assistance duration, age, sex, race, disability status, region, and household composition. A recurring finding in our analysis across subgroups is that although the employment rates and average quarterly earnings of low-income comparators exceed those of entrants both before and after entry, entrants tend to experience post-entry improvements in their labor market trajectories that exceed those of low-income comparators.

Figure 3 displays employment and earnings trajectories by housing assistance program. Across the three main programs, Housing Choice Vouchers, public housing, and project-based Section 8, entrants have employment and earnings trajectories that resemble those in the full sample of entrants: employment and earnings tend to decline before entry and then increase after entry, and these positive changes in trend exceed those of (adjusted) low-income comparators. Entrants in other multifamily programs have levels and trends of employment and earnings trajectories that more closely resemble those of low-income comparators. For this group of entrants, employment and earnings increase in the quarters leading up to entry and then either stabilize or continue to increase after entry, and both changes in trend are statistically indistinguishable from those of low-income comparators.

Figure 4 stratifies the analysis by the duration of HUD assistance, which allows us to examine entrants' employment and earnings trajectories before, during, and after HUD assistance. We present results across six subgroups of entrants defined by their duration in a HUD program: four quarters (one year), eight quarters (two years), twelve quarters (three years), sixteen quarters (four years), twenty quarters (five years), and at least forty quarters (ten years). Across all subgroups, entrants have declining pre-entry employment and earnings trajectories that diverge from those of low-income comparators. Entrants within each subgroup then exhibit positive post-entry changes in their labor market trajectories which exceed those of low-income comparators, though the difference in employment trend breaks is statistically insignificant for those in a HUD program for four quarters. While the sizes of these positive associations are similar for employment, they tend to decrease with duration for earnings. The smaller post-entry increases in earnings among those who receive HUD assistance for relatively longer durations may reflect higher pre-existing levels of need, on average (e.g., due to chronic health issues or other unmeasured barriers to employment that the baseline outcome adjustments fail to capture), and vice-versa.

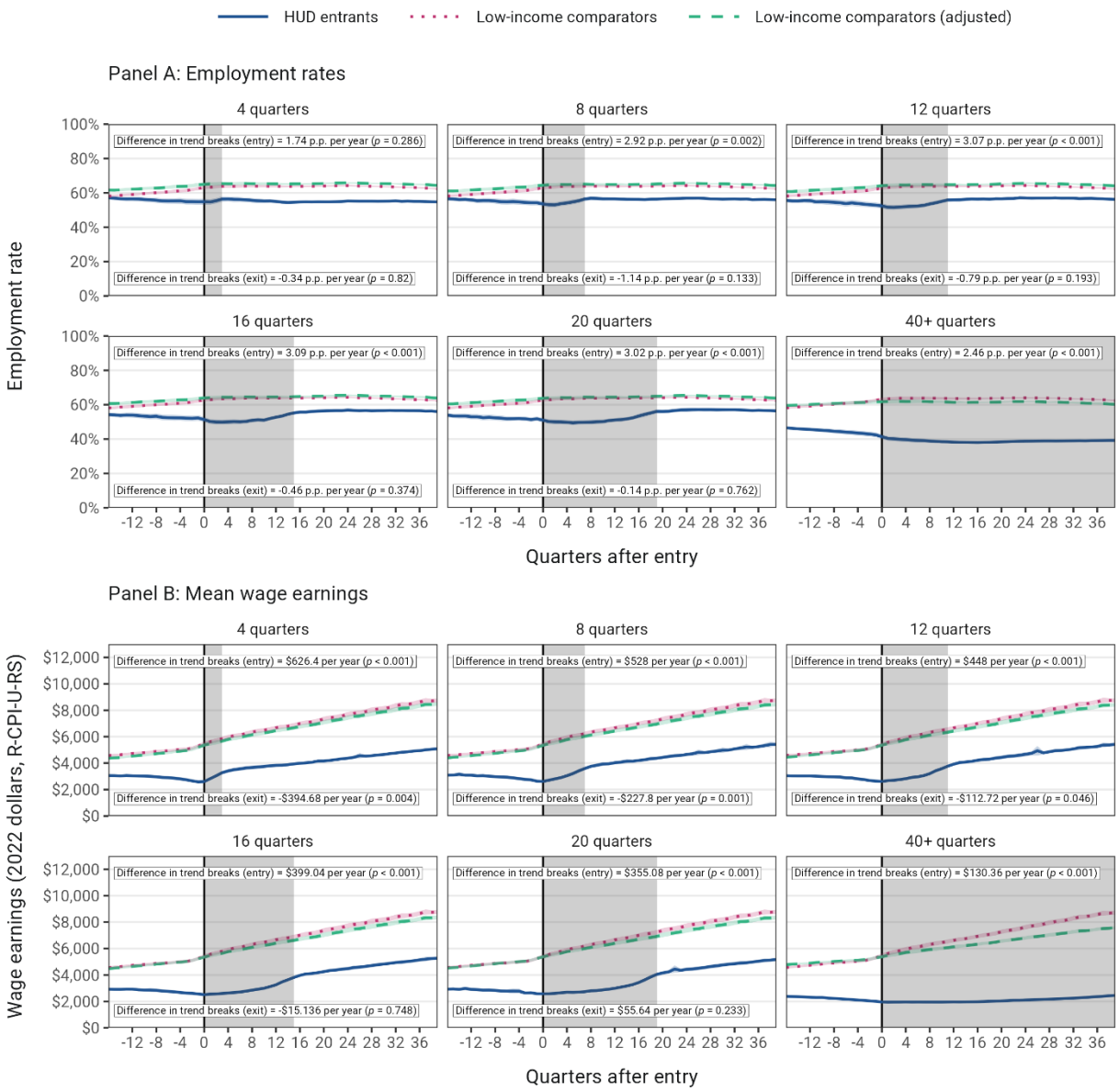
Figure 3: Labor market trajectories of HUD entrants and low-income ACS comparators by HUD program



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and program-specific sub-panel (the same comparators are used in all sub-panels). Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Figure 4: Labor market trajectories of HUD entrants and low-income ACS comparators by assistance duration



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and duration-specific sub-panel (the same comparators are used in all sub-panels). Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts. Gray shaded areas highlight the quarters in which entrants were participating.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Foreshadowing the results of the exit analysis, the slopes of entrants' post-exit trajectories are flatter than the positive slopes observed during their participation. While these negative trend breaks are typically more extreme than those of low-income comparators, most of the estimated differences are statistically insignificant. This lack of precision may reflect the relatively smaller subgroup sample sizes, which we overcome in the exit analysis by considering all HUD participants who exited a HUD program between 2005 and 2011.

The estimates in Figure 5 indicate that although the levels and trends of entrants' labor market trajectories vary substantially with the age at which they entered a HUD program, entry is associated with improved labor market trajectories within each age group. For example, individuals who enter HUD under the age of 35 have stable pre-entry employment and earnings trajectories that then increase to higher-than-baseline levels in the 10 years after entry. Individuals who enter HUD at older ages, on the other hand, experience declining pre-entry employment and earnings trajectories that then either stabilize (as they do for those who enter over the age of 44) or reverse (as they do for those who enter between 35 and 44) after entry. Within each age group, however, entrants exhibit positive post-entry changes in trend that exceed those of low-income comparators.

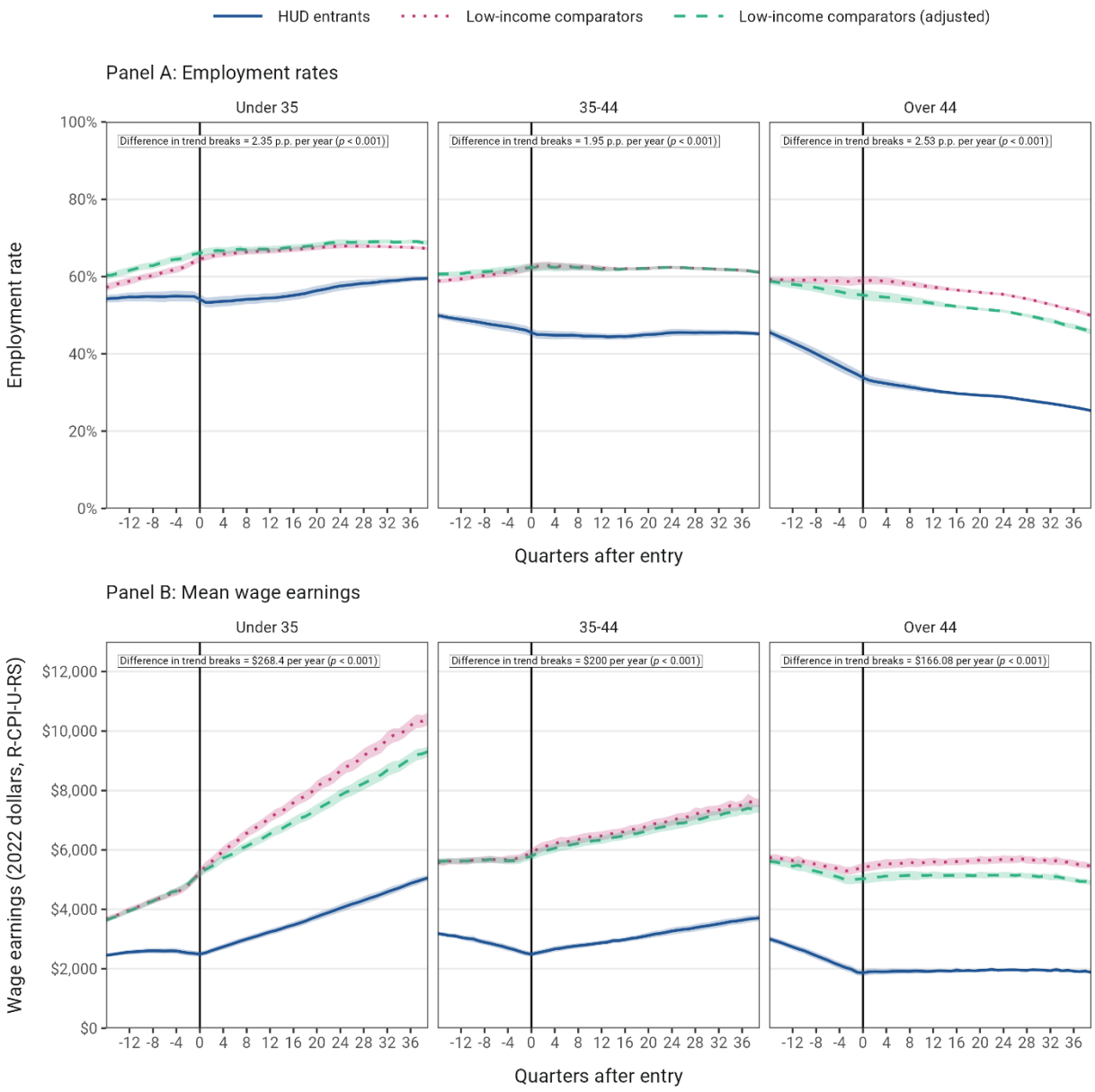
Figure 6 presents labor market trajectories by sex. While both female and male entrants have decreasing employment rates before entry, female entrants' employment rates tend to increase after entry whereas male entrants' employment trajectory tends to flatten. However, the positive-signed difference in post-entry trend breaks between entrants and (adjusted) low-income comparators is similar within each group, indicating that the association of entry with employment trajectories exhibits relatively little heterogeneity by sex. There is relatively more variation in the association of entry with earnings trajectories, which is somewhat larger for female entrants than for male entrants.

Figure 7 presents labor market trajectories for the four largest racial groups: non-Hispanic White (NH White), non-Hispanic Black (NH Black), Hispanic, and non-Hispanic Asian (NH Asian). Pre-entry employment trajectories vary substantially across each group of entrants, as NH White HUD entrants have declining employment rates, NH Black and Hispanic entrants have stable employment rates, and NH Asian entrants have increasing employment rates. All four groups nevertheless experience post-entry trend breaks that exceed those of (adjusted) low-income comparators, consistent with a positive association between entry and employment trajectories, though the difference is relatively smaller for NH Asian entrants. Similarly, all four groups exhibit positive post-entry changes in earnings trends that exceed those of low-income comparators, indicating that entry is associated with improved earnings trajectories. As with employment, the estimated association is relatively smaller for NH Asian entrants.

The estimates in Figure 8 demonstrate that labor market trajectories vary substantially between individuals with and without a disability. Entrants without a disability have pre-entry employment and earnings trajectories that are much flatter than the sharply declining trajectories those of entrants with a disability. Both groups of entrants then experience improved post-entry trajectories, but the stabilization of labor market trajectories among entrants with a disability represents a particularly large improvement relative to their pre-entry trajectories. The improvements for both groups of entrants exceed those of (adjusted) low-income comparators, indicating that entry is associated with improved trajectories within each group. Relative to their low baseline employment rates and earnings, the estimated associations

are especially pronounced for disabled entrants, which suggests that housing may be particularly crucial for this highly vulnerable group facing difficult economic outcomes.

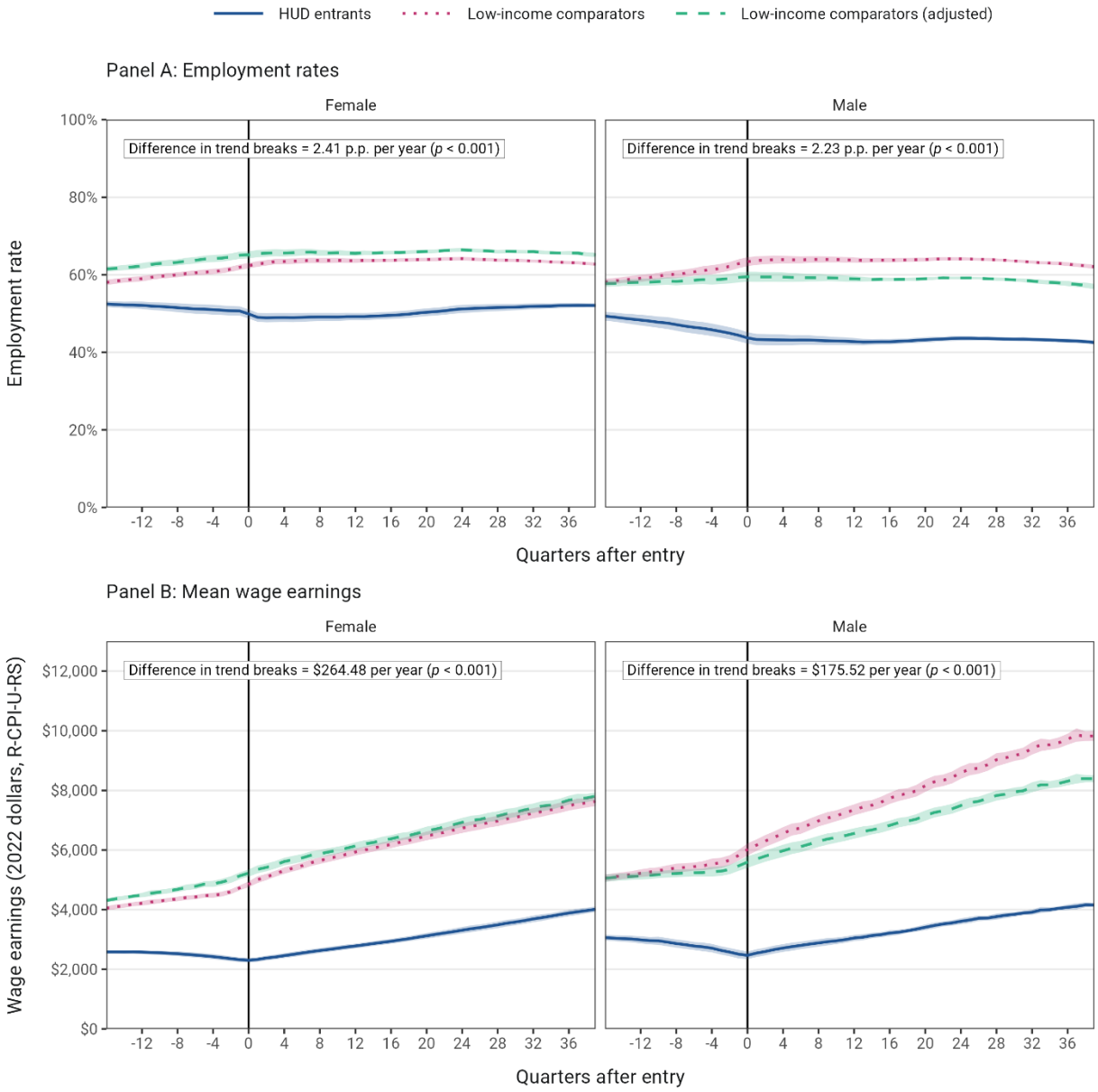
Figure 5: Labor market trajectories of HUD entrants and low-income ACS comparators by age group



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and age group. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

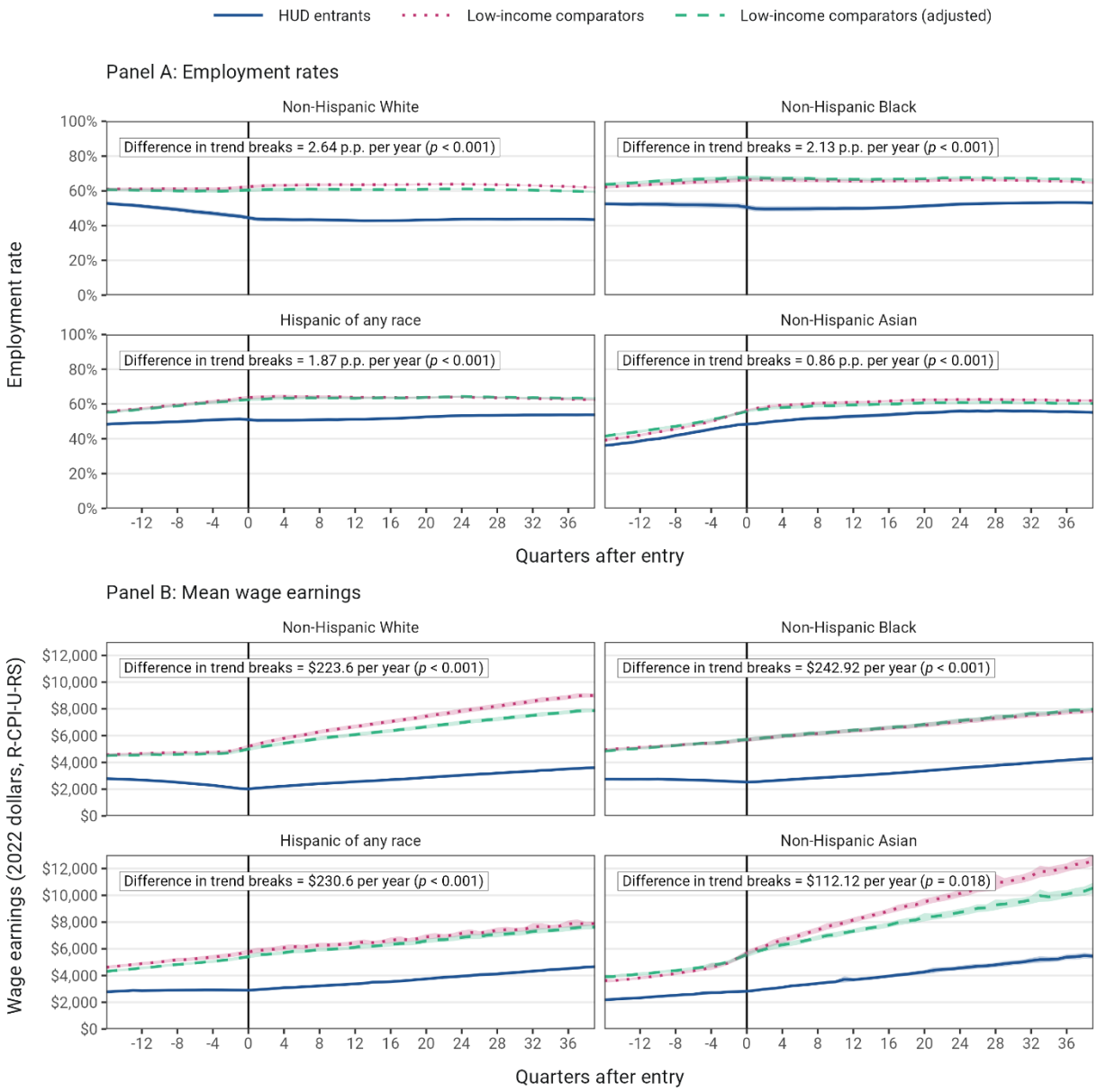
Figure 6: Labor market trajectories of HUD entrants and low-income ACS comparators by sex



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and sex. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

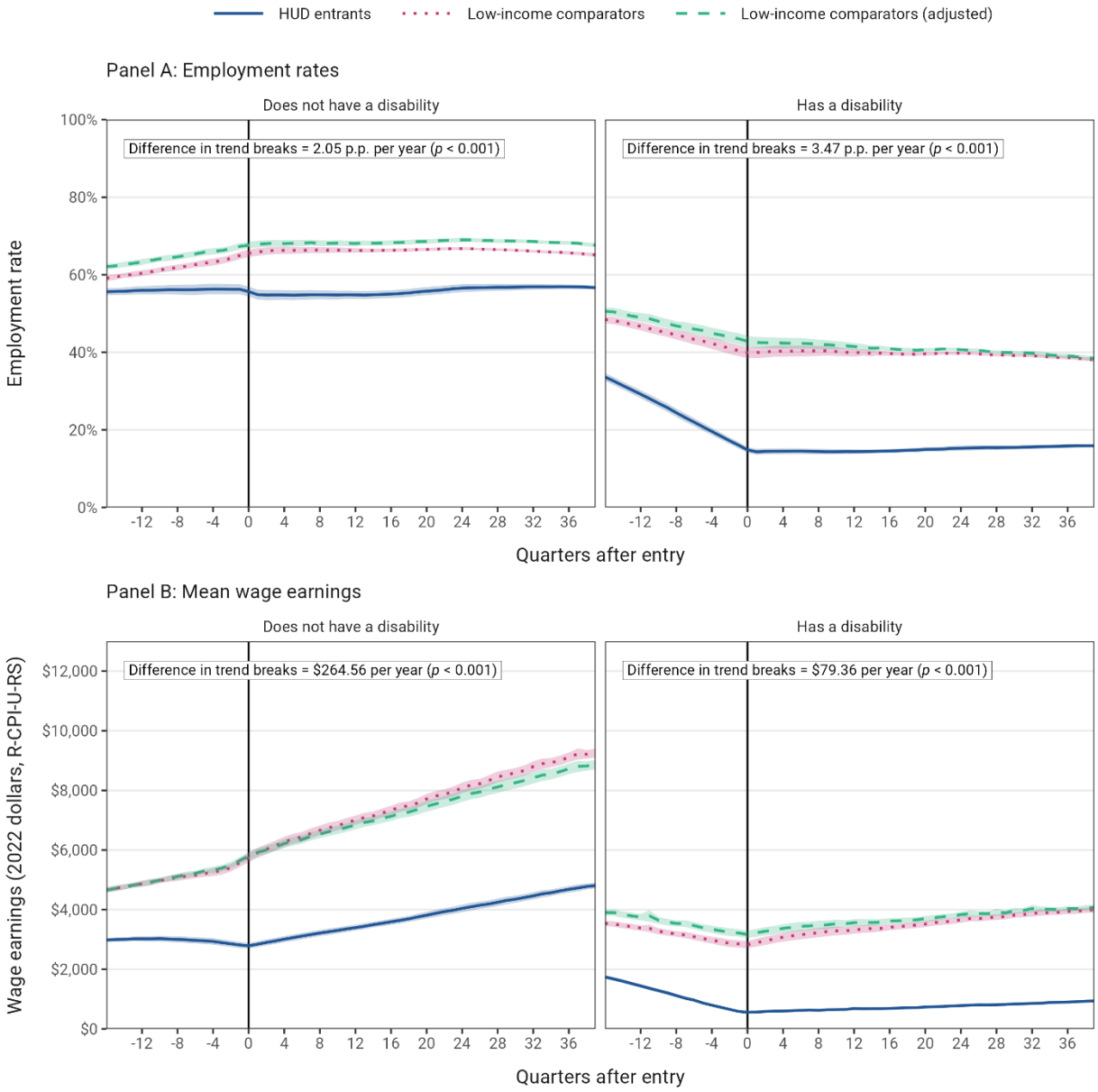
Figure 7: Labor market trajectories of HUD entrants and low-income ACS comparators by race



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and racial category. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

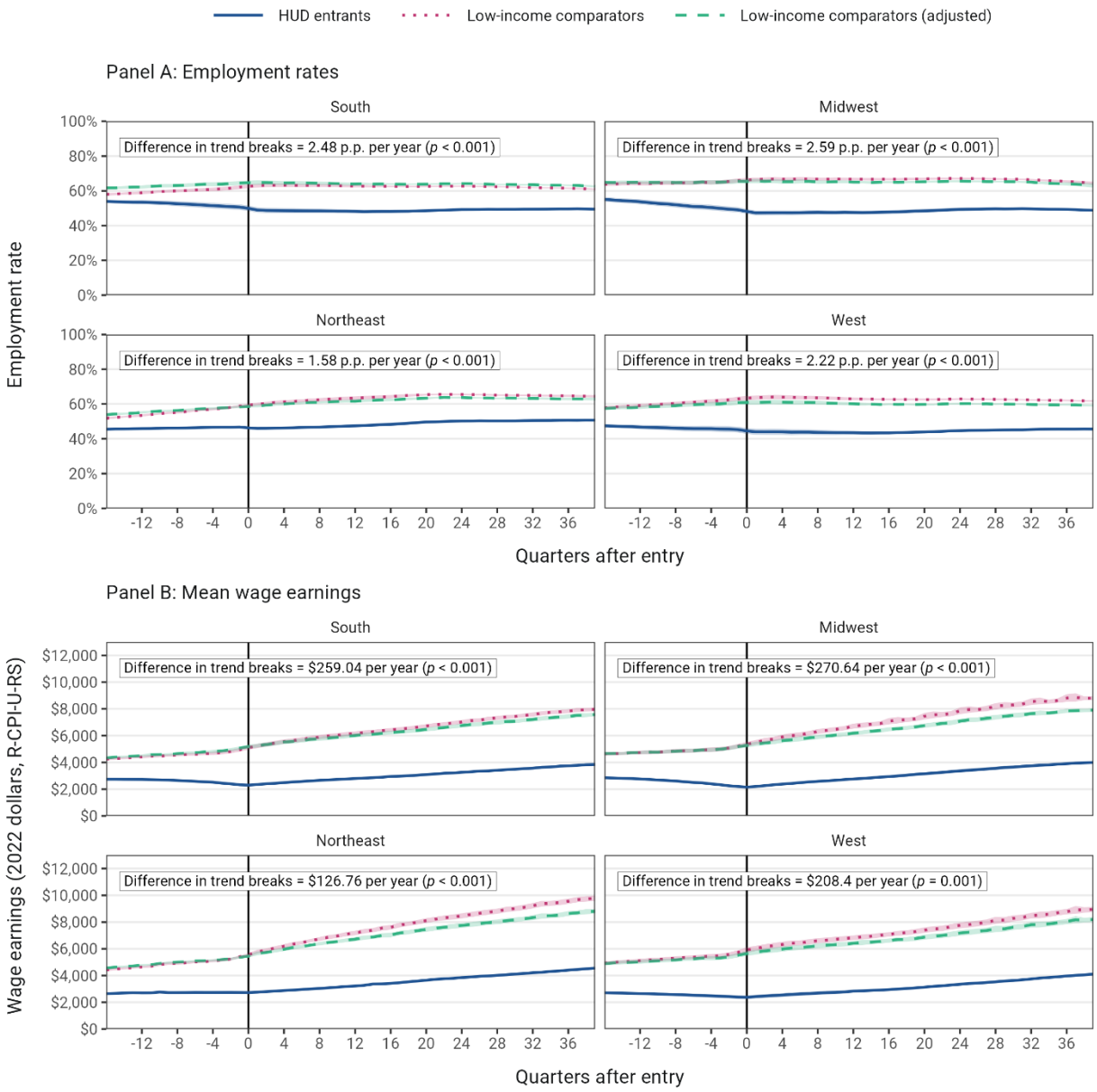
Figure 8: Labor market trajectories of HUD entrants and low-income ACS comparators by disability status



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and disability status. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

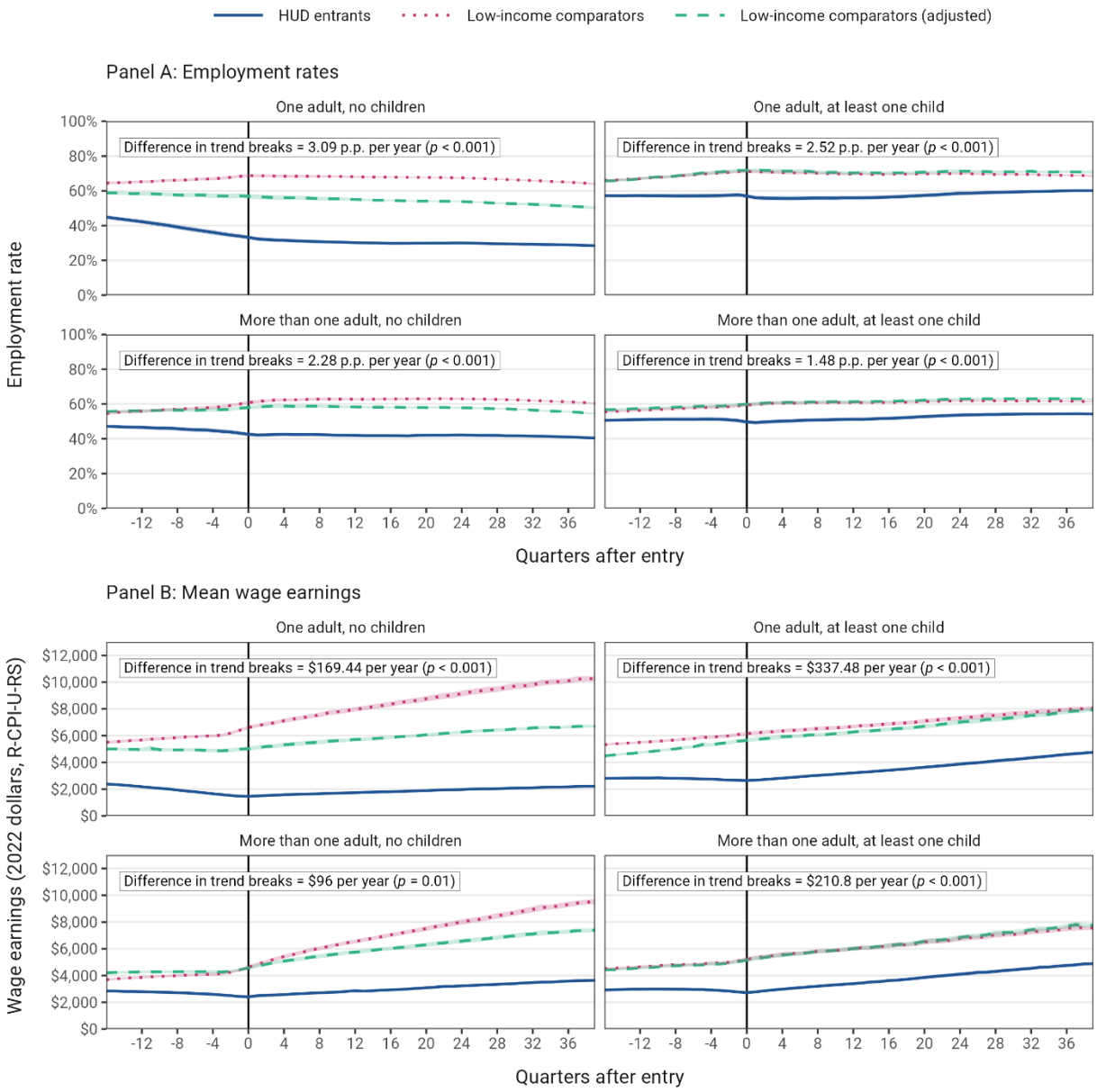
Figure 9: Labor market trajectories of HUD entrants and low-income ACS comparators by region



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and region. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Figure 10: Labor market trajectories of HUD entrants and low-income ACS comparators by household type



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort and household type. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Figure 9 presents labor market trajectories by Census region. While entrants in the South, Midwest, and West regions exhibit employment and earnings trajectories like those of entrants in the full sample, entrants in the Northeast exhibit comparatively flatter pre-entry trajectories. In this way, the post-entry improvements in the labor market trajectories of entrants in the Northeast are less pronounced than those of entrants in other regions. Accordingly, the post-entry increases for entrants in the Northeast exceed those of (adjusted) low-income comparators in the Northeast, but the size of these differences is smaller than those between entrants and low-income comparators in other regions.

Finally, the estimates in Figure 10 indicate that labor market trajectories vary by household composition. For example, entrants in households without children tend to have steeper pre-entry declines in their employment and earnings than entrants in households with children, and their post-entry changes are relatively less pronounced and sometimes opposite signed. Despite these differences, the estimated association for each outcome and household type is positive and statistically significant, though the size of the estimated associations varies.

6.B. Exit Analysis

Full-Sample Trajectories

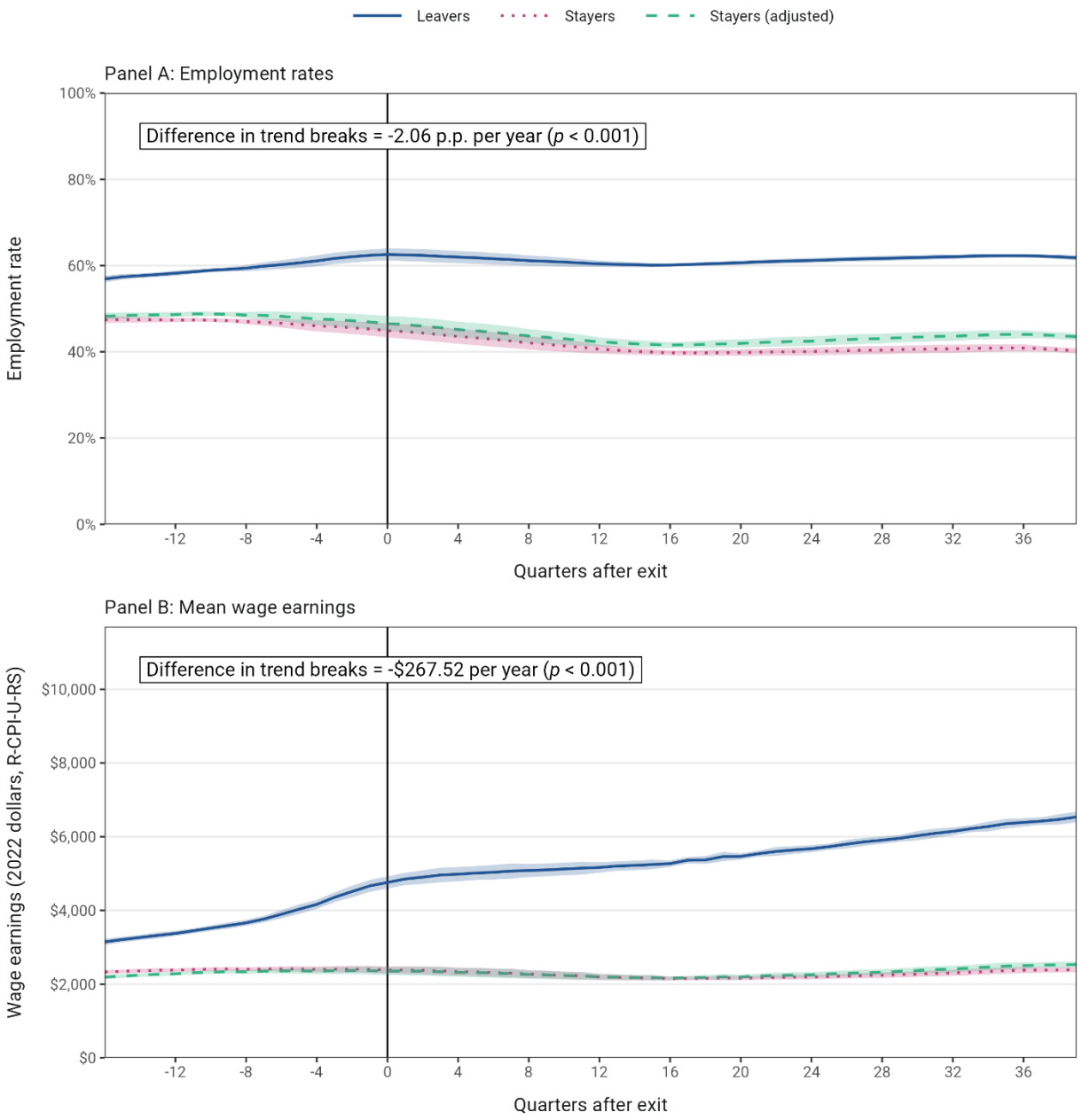
Turning to the exit analysis, we present the labor market trajectories of leavers and stayers in Figure 11. In Panel A we show that prior to exiting a HUD program, leavers' employment rates increase by an average of 1.05 percentage points per year, on average, whereas (adjusted) stayers' employment rates decrease by an average of 1.28 percentage points per year over the same 4-year period. In the 10 years following their exit, leavers exhibit a trend in their employment rates that is statistically indistinguishable from zero. Relative to their pre-exit trend, the slope of leavers' employment trajectories decreases by 1.05 percentage points per year following their exit from a HUD program, which is 2.06 percentage points per year less than the change for stayers. This suggests that exit is associated with diminished post-exit employment trajectories among leavers.

In Panel B we document a similar pattern for earnings. In the quarters leading up their exit, leavers' quarterly earnings increase at an annualized rate of $425 per year, on average, whereas stayers exhibit an earnings trend that is statistically indistinguishable from zero. In the 10 years after exit, we find that leavers' earnings increase by an average of $181 per year, which is $244 per year less than their pre-exit trend. This change in trend is $268 per year less than that of stayers, which suggests that while leavers' earnings continue to increase, exit is nevertheless associated with negative changes in leavers' earnings trajectories.

Group-Specific Trajectories

In Figures 12–18, we present the economic trajectories of leavers and stayers disaggregated by program type, age, sex, race, disability status, region, and household composition. A recurring finding in our analysis across subgroups is that although the employment rates and average quarterly earnings of leavers exceed those of stayers both before and after entry, the slopes of leavers' labor market trajectories tend to diminish after exiting whereas the slopes of stayers' trajectories exhibit smaller changes.

Figure 11: Labor market trajectories of HUD leavers and stayers



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

The estimates in Figure 12 illustrate that the association of exiting a HUD program with labor market trajectories does not vary much across programs. Consistent with the full-sample analysis, leavers from each program have increasing pre-exit employment trajectories that then level off after exiting. Similarly, leavers from each program have increasing pre-exit earnings trajectories that diminish after exiting. These post-entry decreases in the slopes of leavers' labor market trajectories are larger than those of stayers within each program, which suggests that exiting Housing Choice Vouchers, Public Housing, Project-based Section 8, or other multifamily programs is associated with negative changes in leavers' labor market trajectories.

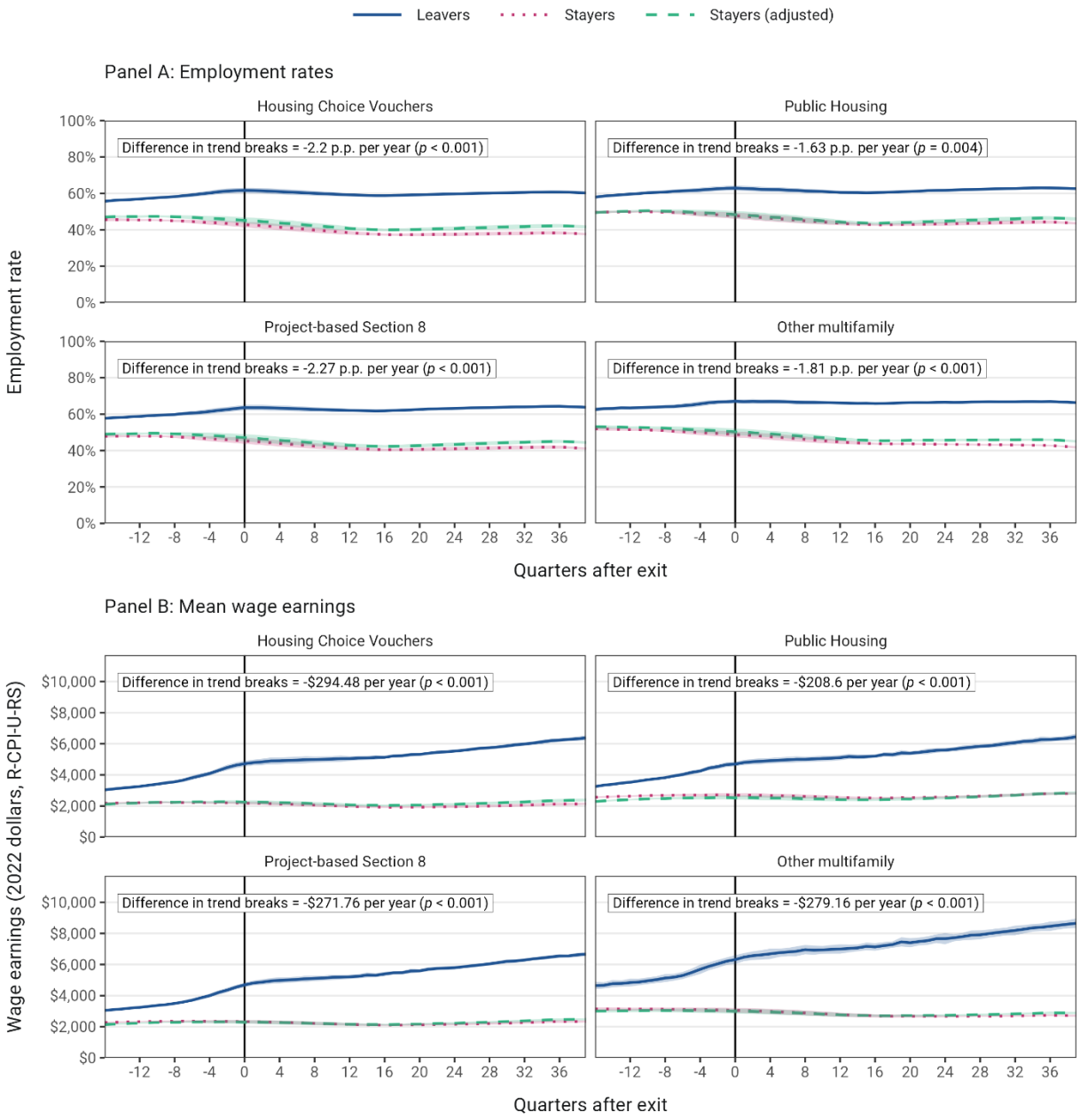
The estimates in Figure 13 indicate that although the levels and trends of leavers' labor market trajectories vary substantially with the age at which they left a HUD program, exiting is associated with diminished labor market trajectories within each age group. For example, individuals who exit over the age of 44 have stable employment trajectories and modestly increasing earnings trajectories before exiting that then become negatively sloped after exiting. Individuals who exit at younger ages exhibit increasing pre-exit employment and earnings trajectories that then diminish in slope after exiting. Within each age group, however, leavers exhibit negative post-entry changes in trend that are more extreme than those of stayers.

Figure 14 stratifies the analysis by sex. Consistent with the full-sample analysis, both female and male leavers have increasing pre-exit employment and earnings trajectories which then diminish in slope after exiting, and these negative post-exit changes in slope are more extreme than those of stayers. In this sense, the estimated association of exiting with subsequent labor market trajectories is negative within each sex, though estimates are somewhat larger in magnitude for female leavers than for male leavers.

Figure 15 illustrates labor market trajectories for the four largest racial groups. All four groups of leavers have increasing employment trajectories before exiting, though the slope for NH White leavers is somewhat flatter than that of NH Black leavers and considerably flatter than that of Hispanic leavers and NH Asian leavers. The employment trajectories of NH Black, Hispanic, and NH Asian leavers then flatten after exiting whereas the trajectories of NH White leavers become somewhat downward sloping. Within all four groups, the negative post-entry change in slope is statistically significantly larger than that of (adjusted) stayers within each group, though this association is smaller for NH White leavers than for other groups of leavers. As in the full sample, all four groups of leavers exhibit increasing earnings trajectories that then diminish after exiting, though the estimated association is less pronounced for NH White leavers.

The estimates in Figure 16 indicate that despite vast differences in employment and earnings levels, leavers with and without a disability exhibit labor market trajectories that are consistent with the full-sample analysis. While leavers with a disability have employment rates and earnings that are typically less than half that of leavers without a disability both before and after exiting, both groups of leavers have increasing pre-exit employment and earnings trajectories which then diminish in slope after exiting. Relative to the stayers within each disability status, leavers experience larger negative changes in the slope of their labor market trajectories after exiting, consistent with a negative association of exiting with subsequent labor market trajectories. The estimated associations are more than twice as large for leavers without a disability than for leavers with a disability, which largely reflects the size of the baseline differences between each group of leavers.

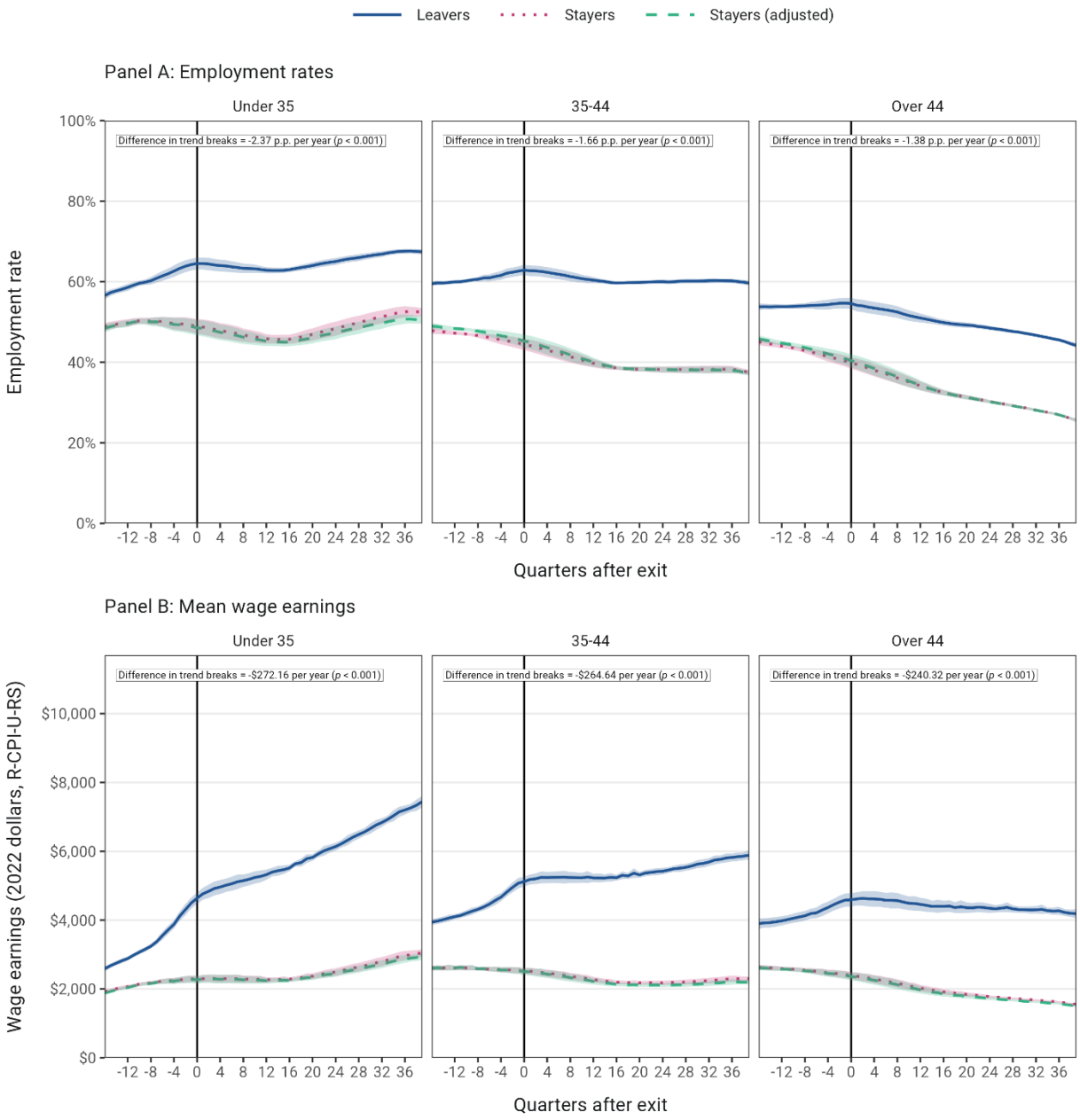
Figure 12: Labor market trajectories of HUD leavers and stayers by HUD program



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort and program. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

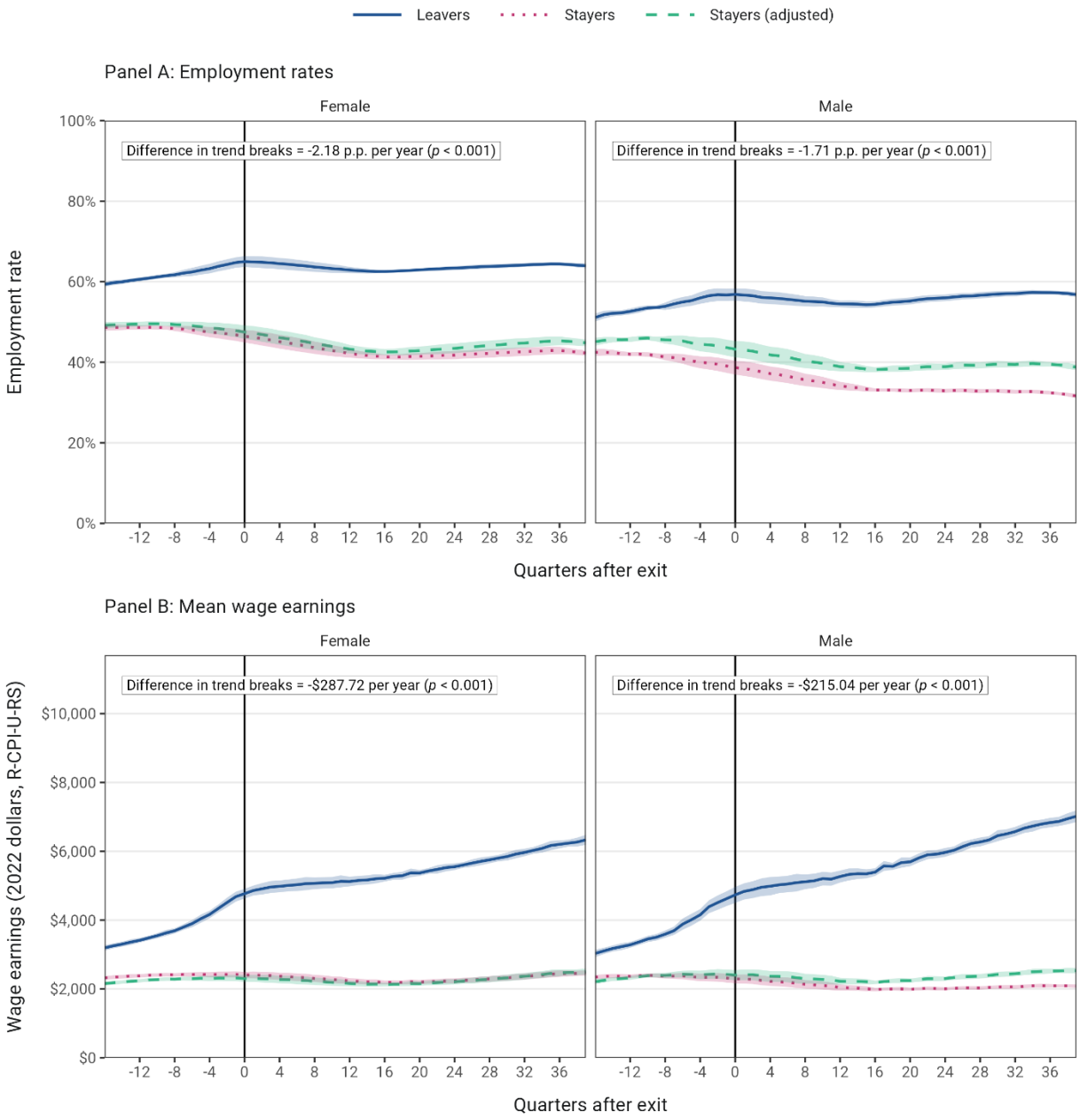
Figure 13: Labor market trajectories of HUD leavers and stayers by age group



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort and age group. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

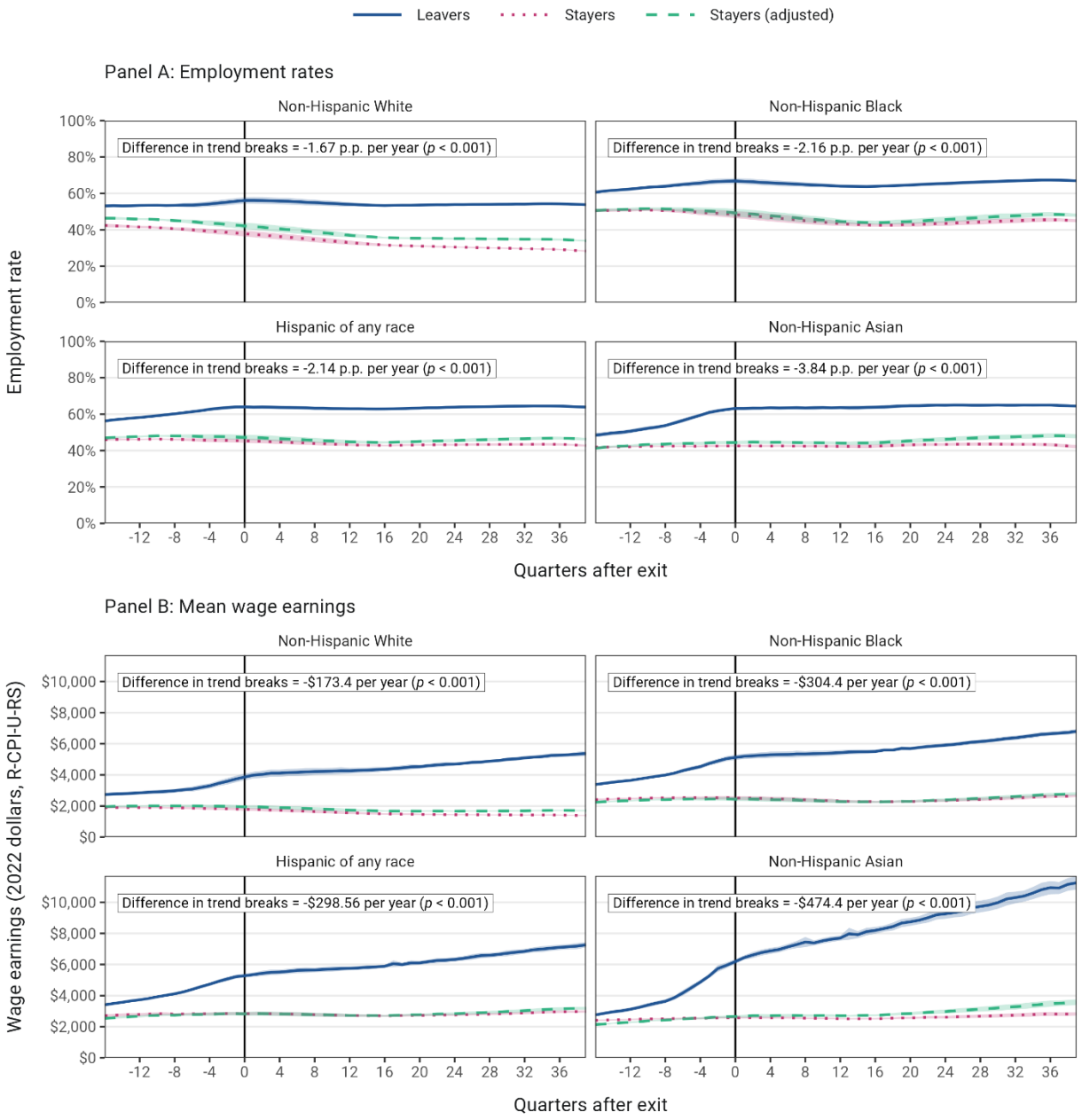
Figure 14: Labor market trajectories of HUD leavers and stayers by sex



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort and sex. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

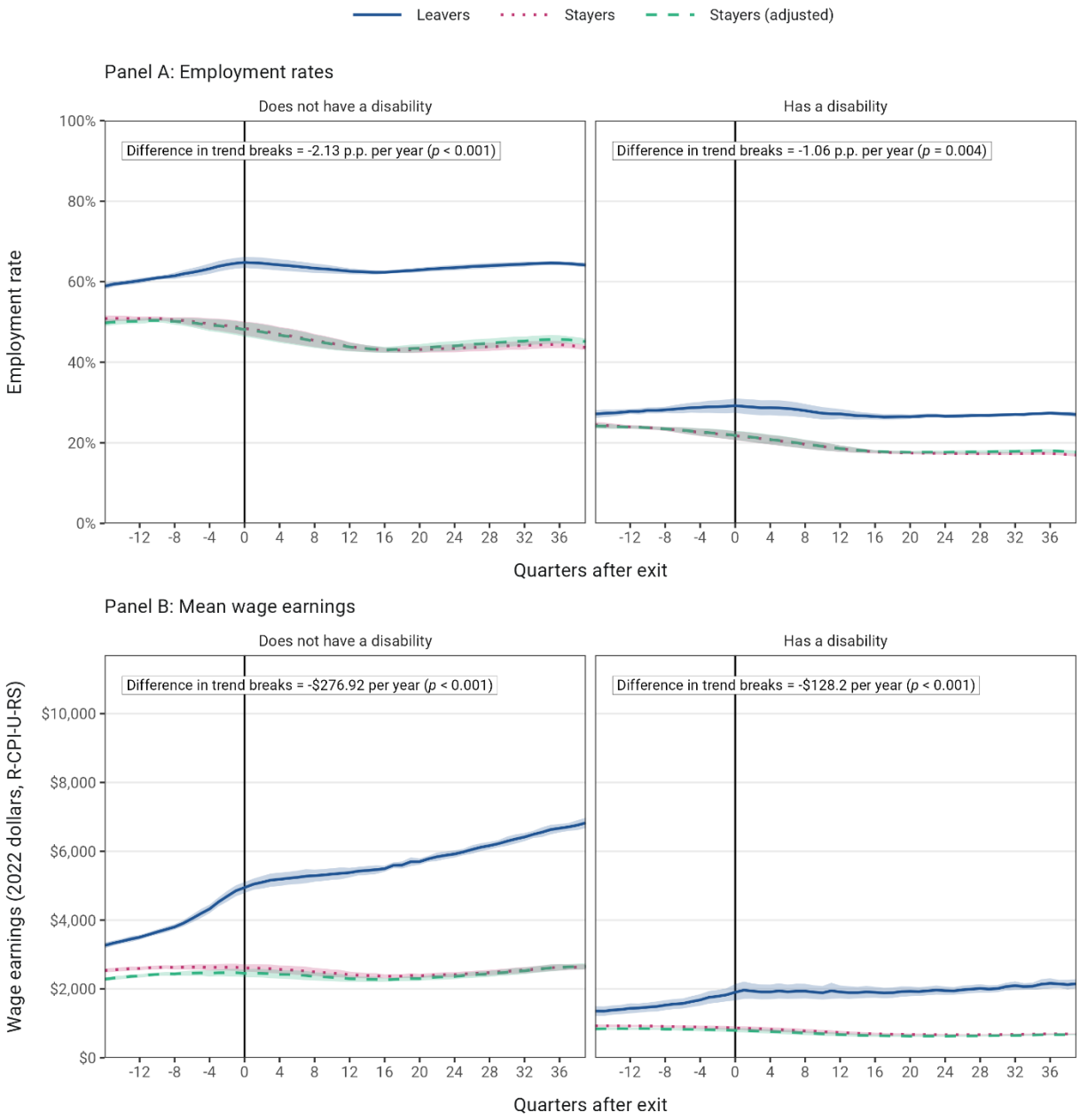
Figure 15: Labor market trajectories of HUD leavers and stayers by race



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort and racial category. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

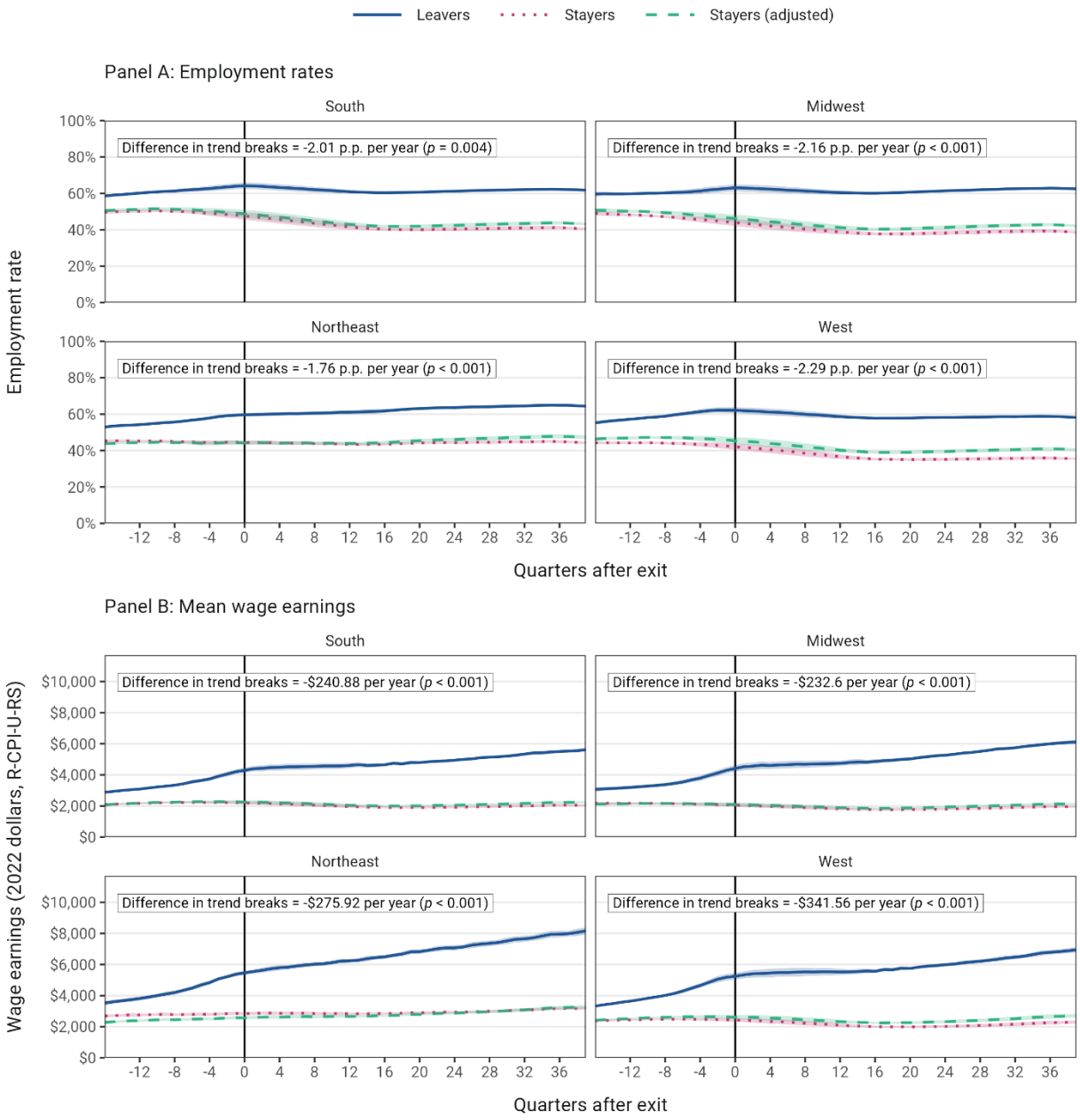
Figure 16: Labor market trajectories of HUD leavers and stayers by disability status



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort and disability status. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

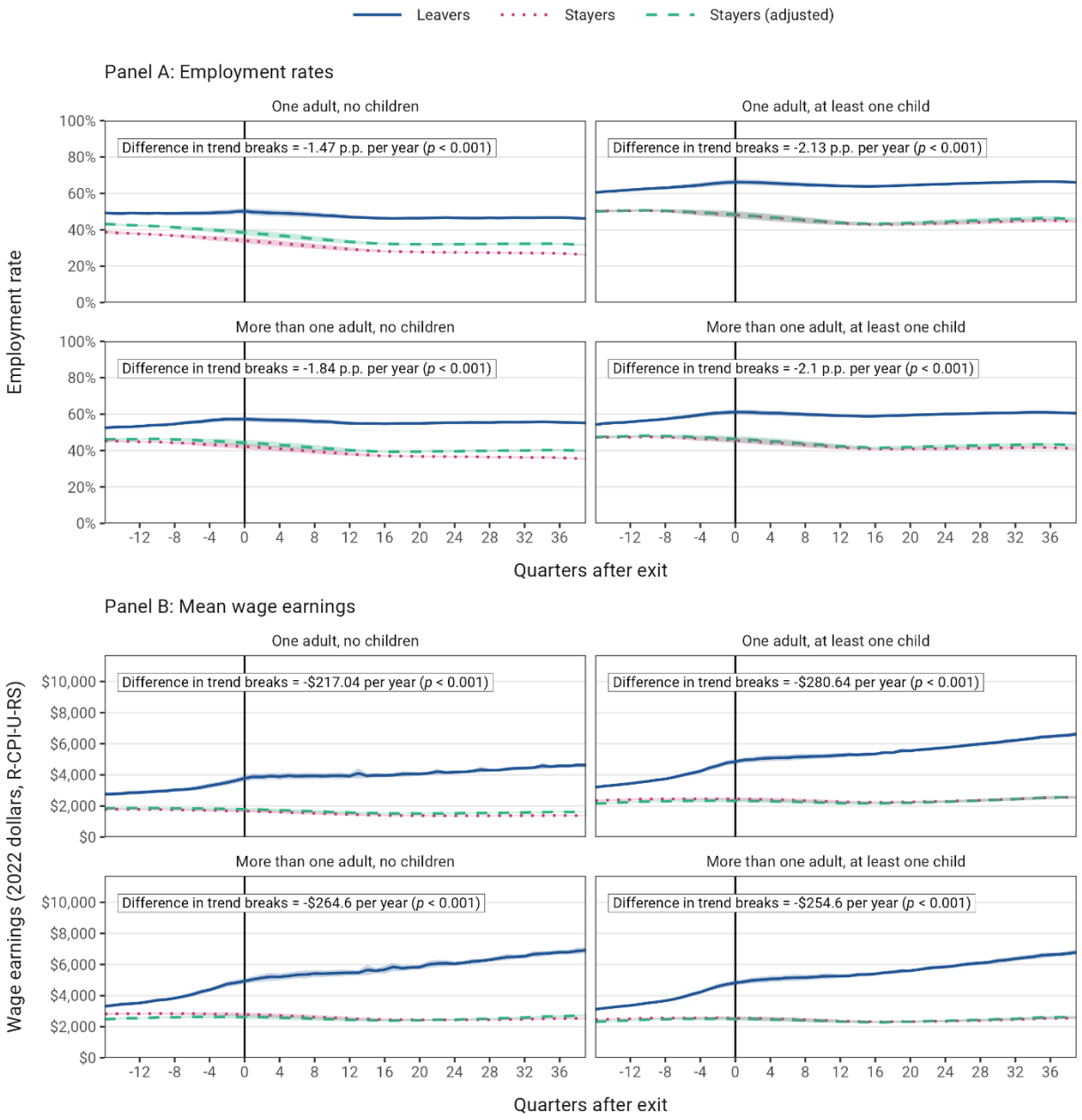
Figure 17: Labor market trajectories of HUD leavers and stayers by region



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort and region. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Figure 18: Labor market trajectories of HUD leavers and stayers by household type



Notes: Leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. Stayers are working-age adults who remained in a HUD program throughout the analysis window. Adjusted trajectories are obtained by reweighting stayers to match the mean non-outcome characteristics of leavers within each quarterly exit cohort and household type. Vertical black lines mark the quarter leavers stopped participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within exit cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Figure 17 presents labor market trajectories by region. Consistent with the full-sample analysis, leavers within each region exhibit increasing pre-entry employment rates and earnings trajectories which then diminish in slope after exiting. Comparing these changes in slope to those of stayers yields negative estimated associations of exiting with subsequent employment and earnings trajectories within each region.

Finally, the estimates in Figure 18 illustrate variation in labor market trajectories by household composition. Leavers within each household type have increasing employment trajectories before exiting, but single adults without children exhibit much flatter pre-exit trajectories. Leavers' employment trajectories then diminish in slope after exiting, becoming flat for leavers in households with children and downward sloping for leavers in households without children. However, leavers experience a larger negative change in slope than stayers within each household type, indicating that exiting is negatively associated with subsequent employment trajectories. Earnings trajectories for each group of leavers resemble those of leavers in the full sample, and the estimated association of exiting with subsequent earnings trajectories are negative for leavers within each household type.

7. Discussion and Conclusions

In this report we examine how employment and earnings evolve for recipients of HUD housing assistance before, during, and after participation. Leveraging the Census Bureau's data infrastructure, we link HUD administrative records to the LEHD to track up to 14 years of quarterly outcomes for the near universe of individuals who either entered a HUD program for the first time between 2005 and 2014 or exited a HUD program for the last time between 2005 and 2011. To estimate the association of entering a HUD program on subsequent labor market trajectories, we construct a comparison group of never-participating low-income renters from the ACS and compare their employment and earnings trajectories to those of HUD entrants. Similarly, to estimate the association of exiting a HUD program on subsequent labor market trajectories, we construct a comparison group of stayers who remain in HUD programs and compare their labor market trajectories to those of leavers. Both analyses span 16 quarters prior to entry or exit and up to 40 quarters afterward, allowing for a longer-run view of economic outcomes. We further disaggregate these trajectories by program type, duration of HUD assistance, age, sex, race, disability status, region, and household composition to provide a richer picture of the HUD population. To our knowledge, this is the most comprehensive population-level examination of HUD recipients' labor market outcomes to date.

In our entry analysis, we find that although HUD entrants have lower employment rates and earnings than their low-income comparators both before and after entry, entrants experience a reversal of their declining pre-entry labor market trajectories after they enter HUD program for the first time, and these post-entry increases in employment and earnings trends exceed those of low-income ACS comparators. When disaggregated by subgroup, the results largely exhibit the same pattern: entrants' labor market trajectories tend to decline before entry and then improve thereafter.

Mirroring the results of our entry analysis, our exit analysis suggests that exiting a HUD program is associated with diminished labor market trajectories. While leavers have higher employment rates and earnings than stayers both before and after exit, their increasing labor market trajectories tend to flatten after they exit a HUD program for the last time. This contrasts

the experience of stayers, who tend to have relatively stable employment and earnings throughout the analysis. Disaggregating the analysis by subgroup produces similar findings.

Together, our longitudinal entry and exit analyses provide suggestive evidence of a positive association of HUD program participation with employment and earnings trajectories. This differs from the main findings of most other longitudinal studies, which tend to document negative (Susin, 2005; Olsen et al., 2005; Carlson et al., 2012; Jacob and Ludwig, 2012) or inconclusive (Van Ryzin et al., 2003) estimates of the association.

The methods we use to estimate the association of HUD program participation with labor market trajectories differ from those used in other non-experimental longitudinal studies. For example, while we employ a difference-in-trend-breaks approach that adjusts for baseline trend differences, Olsen et al. (2005) employ methods akin to difference-in-differences using data with one pre-entry observation for each individual, which prevents them from detecting and adjusting for any baseline differences in employment and earnings trends between HUD entrants and the survey-based comparison group in their analysis. By reweighting only on non-outcome characteristics, we also depart from Susin (2005) and Carlson et al. (2012), who construct their comparison groups by explicitly matching on past labor-market outcomes, a procedure that can lead to regression-to-the-mean bias (Daw and Hatfield, 2018; also acknowledged by Susin, 2005). The way we construct our comparison groups and adjust for baseline trends in our analysis helps reduce the scope for biases from pre-existing trends and regression to the mean, which may explain why our estimates of the association differ from the non-experimental estimates of Susin (2005), Olsen et al. (2005), and Carlson et al. (2012).

The contrast between our non-experimental estimates and the quasi-experimental estimates of Jacob and Ludwig (2012) may reflect both methodological and contextual differences. By leveraging a wait list lottery as a source of as-good-as-random variation in access to vouchers among voucher applicants in Chicago, the research design of Jacob and Ludwig (2012) likely eliminates the scope for selection bias in their analysis. However, their quasi-experimental estimates describe the association of voucher access with labor market outcomes for relatively narrow group of individuals, namely those who entered a wait list lottery in Chicago in 1997. How individuals respond to voucher access could differ in other markets.

We build upon existing studies of individual cities or states (e.g., Jacob and Ludwig, 2012; Carlson et al., 2012; Van Ryzin et al., 2003) by describing the labor market trajectories of HUD program participants throughout the United States. While our non-experimental estimates of the relationship between HUD program participation and labor market trajectories may not fully account for all sources of selection bias,[20] they describe longer-run changes in labor market trajectories for a much broader group of HUD program participants. Taken at face value, our estimates suggest that housing assistance improves the employment and earnings trajectories of HUD recipients over the long run. Future data collection efforts could facilitate nationally representative causal inference on labor supply responses to housing assistance by gathering wait lists and related measures from individual PHAs.

[20] For example, as we show in Appendix D, the subset of HUD entrants who link to the ACS tend to have lower educational attainment, less previous work experience, and fewer significant disabilities than low-income renters who never participate in a HUD program.

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Appendix A: PIK Rates

Table A.2: PIK rates of HUD records

	2005-2022		1995-2004	
	PIK rate	Records	PIK rate	Records
Full sample	98.88%	204,000,000	93.86%	93,790,000
Disability status				
Does not have a disability	98.71%	163,900,000	93.49%	80,810,000
Has a disability	99.58%	40,040,000	97.03%	11,730,000
Missing	72.82%	1,400	88.14%	1,256,000
Sex				
Female	98.93%	127,800,000	94.48%	59,510,000
Male	98.79%	75,980,000	92.81%	34,240,000
Missing	98.47%	193,000	75.01%	39,000
Race				
Non-Hispanic White	99.40%	54,330,000	95.78%	8,457,000
Non-Hispanic Black	99.29%	87,770,000	94.94%	12,560,000
Hispanic of any race	97.76%	41,330,000	93.21%	9,446,000
Non-Hispanic Asian	99.40%	5,052,000	96.37%	601,000
Non-Hispanic two or more races	99.21%	2,395,000	94.69%	269,000
Non-Hispanic American Indian/Alaska Native	99.30%	1,415,000	95.18%	207,000
Non-Hispanic Native Hawaiian/Pacific Islander	99.39%	1,002,000	96.11%	69,000
Non-Hispanic some other race	98.63%	312,000		0
Missing	96.68%	10,370,000	93.44%	62,180,000
Age group				
Under 18	98.72%	78,130,000	91.09%	39,550,000
18-24	99.07%	20,130,000	94.84%	9,544,000
25-34	98.75%	22,010,000	95.87%	10,850,000
35-44	98.30%	18,670,000	96.45%	9,475,000
45-54	98.90%	17,180,000	96.79%	6,340,000
55-64	99.43%	17,280,000	96.69%	4,742,000
Over 64	99.35%	30,560,000	95.76%	13,260,000
Missing	0.00%	25,000	0.24%	36,000
Census region				
South	99.08%	71,080,000	94.14%	33,630,000
Midwest	99.19%	42,050,000	94.42%	20,030,000
Northeast	99.20%	49,280,000	93.64%	20,980,000
West	98.93%	34,720,000	93.89%	16,190,000
Missing	92.20%	6,847,000	88.35%	2,951,000
HUD program				
Housing Choice Vouchers	99.39%	106,800,000	94.30%	41,260,000
Public Housing	99.18%	45,600,000	93.04%	23,930,000
Project-based Section 8	98.49%	45,370,000	94.58%	24,310,000
Other multifamily	98.27%	4,823,000	93.88%	3,747,000
Other		0	90.33%	201,000

Notes: PIK rates represent the percentage of HUD records with a deduplicated PIK. A HUD record represents an annual observation of a person in a HUD-assisted household listed on a HUD-50058, HUD-50058 MTW, or HUD-50059 form.
Data source: HUD PIC/TRACS database. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Table A.2: PIK rates of low-income renters in the ACS

	2005-2022	
	PIK rate	Person records
Full sample	84.70%	13,160,000
Disability status		
Does not have a disability	83.47%	10,500,000
Has a disability	91.03%	2,465,000
Not in universe	83.05%	195,000
Sex		
Female	85.65%	7,226,000
Male	83.59%	5,934,000
Race		
Non-Hispanic White	90.57%	6,196,000
Non-Hispanic Black	87.45%	2,470,000
Hispanic of any race	74.74%	3,204,000
Non-Hispanic Asian	80.48%	615,000
Non-Hispanic two or more races	90.28%	411,000
Non-Hispanic American Indian/Alaska Native	84.90%	191,000
Non-Hispanic Native Hawaiian/Pacific Islander	81.96%	32,000
Non-Hispanic some other race	74.67%	40,500
Age group		
Under 18	84.84%	3,870,000
18-24	83.04%	1,710,000
25-34	81.70%	2,055,000
35-44	81.29%	1,505,000
45-54	85.77%	1,287,000
55-64	89.24%	1,123,000
Over 64	91.71%	1,608,000
Census region		
South	84.57%	4,269,000
Midwest	88.94%	3,012,000
Northeast	85.34%	2,445,000
West	81.40%	3,434,000
Citizenship		
Citizen	88.66%	11,680,000
Not a citizen	59.11%	1,479,000

Notes: PIK rates represent the percentage of ACS person records with a deduplicated PIK.

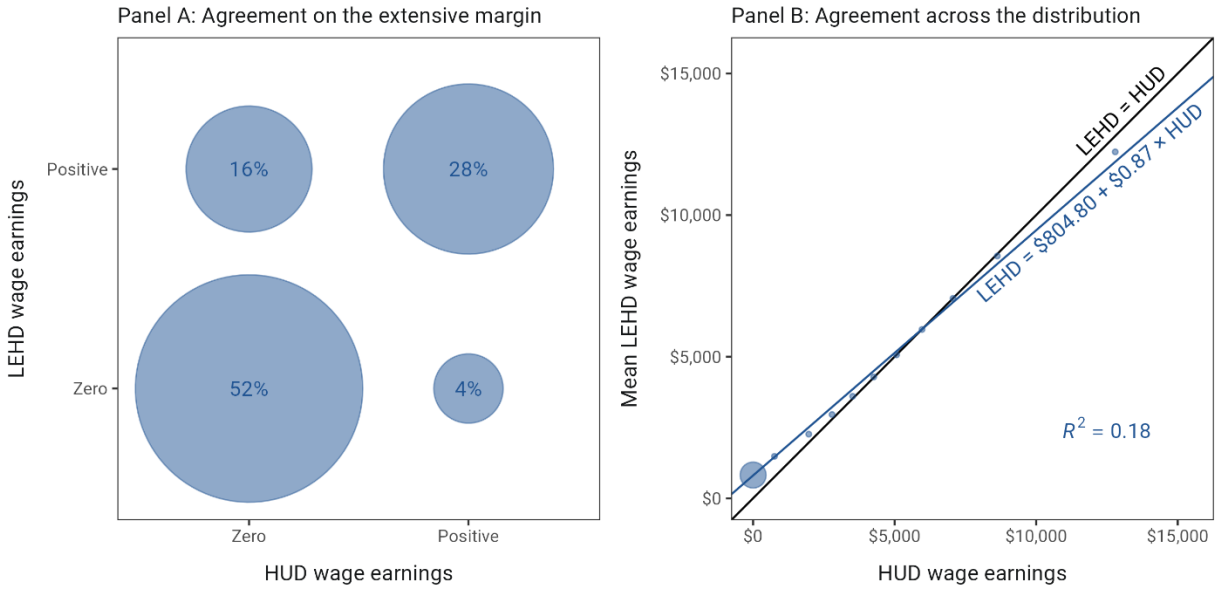
Data source: ACS. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Appendix B: Agreement of Wage Earnings in HUD and LEHD Records

As part of the certification and recertification process, PHAs collect information on the anticipated wage earnings of HUD program participants based on their current employment situation.[21] In this sense, the wage earnings reported in the PIC/TRACS database represent forward-looking projections of future earnings at the time of a certification or recertification transaction. Wage earnings in the LEHD, by contrast, reflect realized earnings.

In Figure B.1, we show that despite this conceptual difference, wage earnings in the LEHD tend to align well, on average, with the wage earnings collected by PHAs as part of the certification and recertification process. For example, the joint distribution in Panel A suggests that wage earnings in the PIC/TRACS database and the LEHD agree 80 percent of the time on the extensive margin, and the binned scatterplot in Panel B shows that positive wage earnings in the PIC/TRACS database tend to correspond with LEHD wage earnings that are nearly equal, on average.

Figure B.1: Agreement of quarterly wage earnings (2022 dollars, R-CPI-U-RS) in HUD and LEHD records



Notes: Statistics in both panels are estimated from a linked dataset that combines individual earnings information from HUD (re)certification transactions and the LEHD database for working-age HUD program participants. The sample consists of 68,020,000 linked transactions between 2005 and 2022. HUD wage earnings represent the amount of money a program participant expected to earn based on their employment situation at the time of the transaction. LEHD wage earnings represent the amount of money a program participant actually earned during the transaction quarter. Panel A shows the joint distribution of transactions across HUD wage earnings and LEHD wage earnings on the zero-nonzero extensive margin. Panel B includes a binned scatterplot (Cattaneo et al., 2024) with ten (roughly equally-sized) bins for transactions that report positive HUD earnings and one (much larger) bin for transactions that report zero HUD wage earnings.

Data sources: HUD PIC/TRACS, LEHD, and the Census Numident. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

[21] See, for example, Section 4 of HUD's Income Determination guide for public housing programs at https://www.hud.gov/sites/dfiles/PIH/documents/PHOG_Income_Determination_FINAL.pdf

Appendix C: LEHD Match Rates

Table C.3: LEHD match rates of HUD entrants and ACS comparators

	HUD entrants		Low-income ACS comparators	
	Match rate	Observations	Match rate	Observations
Full sample	95.31%	2,189,000	93.57%	1,880,000
Disability status				
Does not have a disability	98.00%	1,718,000	94.09%	1,657,000
Has a disability	85.51%	471,000	89.58%	223,000
Sex				
Female	95.62%	1,435,000	93.51%	981,000
Male	94.72%	754,000	93.54%	899,000
Race				
Non-Hispanic White	94.80%	947,000	97.07%	1,090,000
Non-Hispanic Black	96.35%	741,000	96.89%	237,000
Hispanic of any race	94.95%	354,000	87.02%	377,000
Non-Hispanic Asian	92.00%	56,500	92.72%	111,000
Non-Hispanic two or more races	95.46%	50,000	96.84%	36,500
Non-Hispanic AIAN	96.50%	28,000	96.73%	21,500
Non-Hispanic other race	94.24%	12,500	93.12%	8,600
Age group				
Under 35	98.16%	1,168,000	94.62%	981,000
35-44	94.84%	524,000	92.52%	490,000
Over 44	89.12%	497,000	91.90%	410,000
Census region				
South	95.82%	812,000	93.99%	587,000
Midwest	96.35%	538,000	95.11%	445,000
Northeast	92.07%	431,000	93.26%	325,000
West	96.35%	408,000	92.43%	524,000
Household structure				
One adult, no children	91.31%	537,000	96.75%	405,000
One adult, at least one child	98.43%	807,000	97.03%	170,000
More than one adult, no children	91.76%	267,000	94.40%	533,000
More than one adult, at least one child	96.32%	579,000	90.95%	772,000
HUD program				
Housing Choice Vouchers	95.13%	1,093,000		
Public housing	95.04%	638,000		
Project-based Section 8	94.03%	538,000		
Other multifamily	92.83%	78,000		

Notes: Match rates represent the percentage of HUD entrants and ACS comparators who appear in the LEHD database with positive earnings at least once by 2023 Q4. HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Low-income comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 and 2022. AIAN = American Indian/Alaska Native.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Table C.2: LEHD match rates of HUD leavers and stayers

	HUD leavers		HUD stayers	
	Match rate	Observations	Match rate	Observations
Full sample	96.20%	1,282,000	89.87%	33,130,000
Disability status				
Does not have a disability	97.84%	1,182,000	94.01%	27,540,000
Has a disability	76.76%	100,000	69.42%	5,586,000
Sex				
Female	96.64%	897,000	91.05%	26,340,000
Male	95.17%	385,000	85.25%	6,792,000
Race				
Non-Hispanic White	94.50%	403,000	84.66%	8,373,000
Non-Hispanic Black	97.33%	563,000	93.48%	16,980,000
Hispanic of any race	96.59%	229,000	87.35%	5,869,000
Non-Hispanic Asian	94.88%	37,500	84.09%	738,000
Non-Hispanic two or more races	96.48%	29,000	90.42%	748,000
Non-Hispanic AIAN	96.21%	14,500	91.98%	289,000
Non-Hispanic other race	96.81%	5,800	91.85%	136,000
Age group				
Under 35	98.55%	728,000	96.53%	12,780,000
35-44	95.99%	343,000	90.43%	9,711,000
Over 44	88.45%	211,000	81.28%	10,640,000
Census region				
South	96.30%	476,000	91.74%	10,690,000
Midwest	96.71%	278,000	91.35%	7,035,000
Northeast	94.80%	282,000	85.31%	9,676,000
West	97.03%	247,000	92.21%	5,729,000
Household structure				
One adult, no children	88.81%	115,000	80.86%	5,388,000
One adult, at least one child	97.98%	655,000	93.88%	16,340,000
More than one adult, no children	92.58%	106,000	83.57%	3,204,000
More than one adult, at least one child	96.35%	399,000	90.14%	8,070,000
HUD program				
Housing Choice Vouchers	96.11%	639,000	89.60%	15,720,000
Public housing	96.10%	318,000	90.17%	9,440,000
Project-based Section 8	96.52%	278,000	90.11%	6,988,000
Other multifamily	96.20%	45,000	89.61%	959,000

Notes: Match rates represent the percentage of HUD leavers and stayers who appear in the LEHD database with positive earnings at least once by 2023 Q4. HUD leavers are working-age adults who exited a HUD program for the last time between 2005 Q1 and 2011 Q1. HUD stayers are working-age adults who remained in a HUD program throughout the analysis. Stayers can serve as comparators for multiple cohorts of leavers, and the inclusion criteria leads most to appear in the sample more than once (see the Data and Methods section for details). AIAN = American Indian/Alaska Native.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Appendix D: Supplemental Summary Statistics

Due to data limitations, we cannot adjust the weights of low-income comparators for several baseline characteristics that may affect both selection into HUD programs and subsequent labor market trajectories. These characteristics are either unavailable in HUD administrative records or not measured consistently across HUD and ACS data. Potentially important factors include educational attainment, occupational background, and more detailed indicators of household structure and disability. To provide context, this appendix reports supplemental summary statistics on low-income renters from the ACS, highlighting compositional differences between HUD entrants and comparators that we cannot incorporate directly into the entry analysis.

In Table D.1, we show that low-income renters who eventually enter a HUD program differ from low-income renters who never enter a HUD program on each of those omitted characteristics. For example, eventual HUD entrants are significantly less likely to live with a spouse or a partner and are more likely to have a child with an identified disability, both of which could explain part of the pre-entry differences in employment in Table 1 (Chaudry 2006). Eventual HUD entrants also tend to have lower educational attainment, which could explain pre-entry differences in both employment and earnings. Furthermore, differences in more detailed measures of disability suggest that HUD entrants are more likely to experience serious disabilities, as eventual HUD entrants are more likely to be blind or Deaf or hard of hearing and have higher mean scores on measures of difficulty engaging in Activities of Daily Living (ADLs) and Instrumental Activities of Daily Living (IADLs). Conditional on having an identified disability, eventual HUD entrants have higher ADL and IADL scores, on average, than low-income comparators, though they are less likely to be blind or Deaf or hard of hearing.

Table D.4: Characteristics of eventual entrants and other low-income renters in the ACS

	Eventual HUD entrants	Low-income comparators	Difference
Family			
Spouse or partner present	33.01%	41.77%	-8.76 p.p. *
At least one disabled child	5.72%	3.71%	2.01 p.p. *
Observations	180,000	4,921,000	5,101,000
Unique PIKs	174,000	4,759,000	4,933,000
Education			
Less than high school	28.39%	20.32%	8.07 p.p. *
High school	37.25%	30.19%	7.06 p.p. *
Some college or Associate's degree	28.44%	32.83%	-4.39 p.p. *
Bachelor's degree or more	5.92%	16.65%	-10.73 p.p. *
Currently enrolled in school	10.51%	18.24%	-7.72 p.p. *
Observations	180,000	4,921,000	5,101,000
Unique PIKs	174,000	4,759,000	4,933,000
Detailed disability			
Deaf or hard of hearing	4.21%	1.99%	2.22 p.p. *
Blind	5.56%	2.49%	3.07 p.p. *
IADL disability score (0-2)	0.26	0.10	0.16 *
ADL disability score (0-2)	0.23	0.08	0.15 *
Observations	127,000	4,210,000	4,337,000
Unique PIKs	124,000	4,089,000	4,213,000
Detailed disability among those with a disability			
Deaf or hard of hearing	14.32%	16.06%	-1.74 p.p. *
Blind	18.90%	20.11%	-1.21 p.p. *
IADL disability score (0-2)	0.90	0.83	0.07 *
ADL disability score (0-2)	0.77	0.66	0.11 *
Observations	42,500	617,000	660,000
Unique PIKs	42,000	604,000	646,000
Occupations			
Management, business, science, and arts	7.53%	16.59%	-9.06 p.p. *
Service	26.36%	23.29%	3.07 p.p. *
Sales and office	20.02%	20.33%	-0.31 p.p. *
Natural resources, construction, and maintenance	5.18%	9.36%	-4.18 p.p. *
Production, transportation, and material moving	12.92%	13.38%	-0.46 p.p. *
Military	0.08%	0.32%	-0.24 p.p. *
No work experience in the last 5 years	27.90%	16.72%	11.17 p.p. *
Observations	180,000	4,921,000	5,101,000
Unique PIKs	174,000	4,759,000	4,933,000

Notes: Eventual HUD entrants are working-age adults who entered a HUD program by the end of 2022 but had not yet entered when they responded to the ACS between 2005 and 2022. Low-income comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's low-income eligibility limit (set at 80% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 and 2022. Detailed disability information was not available before 2008, so the detailed disability statistics were derived from the 2008-2022 ACS rather than the full 2005-2022 ACS. Commuting statistics were derived from individuals who worked the week before they were interviewed, and commuting-time statistics exclude those who were working at home. Asterisks (*) indicate that an estimated difference is statistically significant at the 5-percent level, based on a statistical test that accounts for heteroskedasticity and clustering at the PIK level. IADL = Instrumental Activities of Daily Living. ADL = Activities of Daily Living.

Data sources: HUD PIC/TRACS, ACS, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Appendix E: Estimates from Alternative ACS Comparison Groups

Balancing the empirical tradeoff between selecting a high-need comparison group and minimizing regression-to-the-mean bias, our main analysis restricts the ACS to renters with incomes at or below the HUD "low-income" limit (80 percent of AMI). In this appendix, we assess alternative restrictions using the more stringent "very low-income" (50 percent of AMI) and "extremely low-income" (30 percent of AMI) thresholds. These alternatives produce comparison groups with baseline outcomes that more closely resemble those of HUD entrants but also display employment and earnings trajectories that are more susceptible to regression-to-the-mean bias.

In Table E.1, we show that very low-income comparators exhibit higher baseline employment rates and higher baseline quarterly earnings, on average, than HUD entrants, and that these differences persist after adjusting for other baseline characteristics. However, the baseline differences in outcomes are much smaller than those between HUD entrants and low-income comparators in Table 1. In Table E.2, we show that extremely low-income comparators have baseline outcomes that are even closer to those of HUD entrants, on average.

However, in Figures E.1 and E.2, we show that the labor market trajectories of these alternative comparison groups, especially those of extremely low-income comparators, exhibit a rebound pattern in which wage earnings reach a nadir around 2-3 quarters before entry and increase steeply thereafter. The timing of this nadir coincides with the reference period for the ACS household income question, which asks respondents about all sources of household income during the 12 months before their survey response, roughly 3-4 quarters before entry depending on the month of the response. Given that wage earnings are an important component of household income, the nadir and subsequent rebound of quarterly wage earnings during this period suggests that the more stringent sample restrictions yield comparison groups with relatively higher shares of individuals experiencing temporary setbacks in the labor market from which they then recover.

In contrast, the low-income comparators in Figure 2 do not exhibit a nadir in their quarterly wage earnings during the 12-month reference period. While this does not rule out the existence of regression-to-the-mean bias among low-income comparators—indeed, there is an inflection point in which earnings begin to increase more steeply during this period—the relatively smoother pre-entry outcome trajectories suggest that this comparison group is less heavily selected on individuals experiencing temporary setbacks. As a result, the outcome trajectories of low-income comparators may better reflect the broader macroeconomic trends faced by HUD entrants, who face persistent economic challenges, than those of the more stringently selected very low-income and extremely low-income comparison groups.

Table E.5: Baseline characteristics of HUD entrants and very low-income ACS comparators

	HUD entrants	Very low-income ACS comparators		Difference (HUD - ACS)	
	Mean	Mean	Adjusted mean	Unadjusted	Adjusted
Baseline outcomes					
Employment rate	50.06%	53.95%	56.06%	-3.89 p.p. *	-6.00 p.p. *
Quarterly wage earnings	$2,620	$3,869	$3,868	-$1,249 *	-$1,248 *
Disability status					
Does not have a disability	80.71%	88.13%	80.71%	-7.42 p.p. *	0.00 p.p.
Has a disability	19.29%	11.87%	19.29%	7.42 p.p. *	0.00 p.p.
Sex					
Female	65.76%	50.96%	65.76%	14.80 p.p. *	0.00 p.p.
Male	34.24%	49.04%	34.24%	-14.80 p.p. *	0.00 p.p.
Race					
Non-Hispanic white	43.04%	45.62%	43.04%	-2.58 p.p. *	0.00 p.p.
Non-Hispanic Black	34.22%	15.20%	34.22%	19.02 p.p. *	0.00 p.p.
Hispanic of any race	16.10%	29.74%	16.10%	-13.63 p.p. *	0.00 p.p.
Non-Hispanic Asian	2.49%	6.41%	2.49%	-3.92 p.p. *	0.00 p.p.
Non-Hispanic two or more races	2.28%	1.61%	2.28%	0.67 p.p. *	0.00 p.p.
Non-Hispanic AIAN	1.29%	0.84%	1.29%	0.44 p.p. *	0.00 p.p.
Non-Hispanic other race	0.57%	0.57%	0.57%	0.00 p.p.	0.00 p.p.
Age					
Age at entry or interview (years)	34.75	34.54	34.75	0.21 *	-0.00
Census region					
South	37.30%	33.83%	37.30%	3.47 p.p. *	0.00 p.p.
Midwest	24.82%	19.35%	24.82%	5.47 p.p. *	0.00 p.p.
Northeast	19.03%	17.64%	19.03%	1.39 p.p. *	0.00 p.p.
West	18.85%	29.18%	18.85%	-10.33 p.p. *	0.00 p.p.
Household characteristics					
Household income	$19,160	$25,260	$22,270	-$6,097 *	-$3,114 *
Number of adults	1.45	1.92	1.45	-0.47 *	-0.00
Number of young children (0-4)	0.53	0.39	0.53	0.14 *	-0.00
Number of children (5-12)	0.58	0.50	0.58	0.08 *	0.00
Number of older children (13-17)	0.22	0.24	0.22	-0.02 *	-0.00
HUD program					
Housing Choice Vouchers	46.81%				
Public housing	27.22%				
Project-based Section 8	22.72%				
Other multifamily	3.26%				
Sample sizes					
Observations	2,086,000	965,000	965,000	3,051,000	3,051,000
Unique PIKs	2,086,000	954,000	954,000	3,040,000	3,040,000

Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Very low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's very low-income eligibility limit (set at 50% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Baseline outcomes are measured during the 16 quarters leading up to the quarter of entry or interview. All dollar amounts are adjusted to 2022 levels using the R-CPI-U-RS. Adjusted statistics are obtained by reweighting comparators to match the mean non-outcome characteristics of HUD entrants within each quarterly entry cohort. Asterisks (*) indicate that an estimated difference is statistically significant at the 5-percent level, based on a statistical test that accounts for heteroskedasticity and clustering at the PIK level. AIAN = American Indian/Alaska Native.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

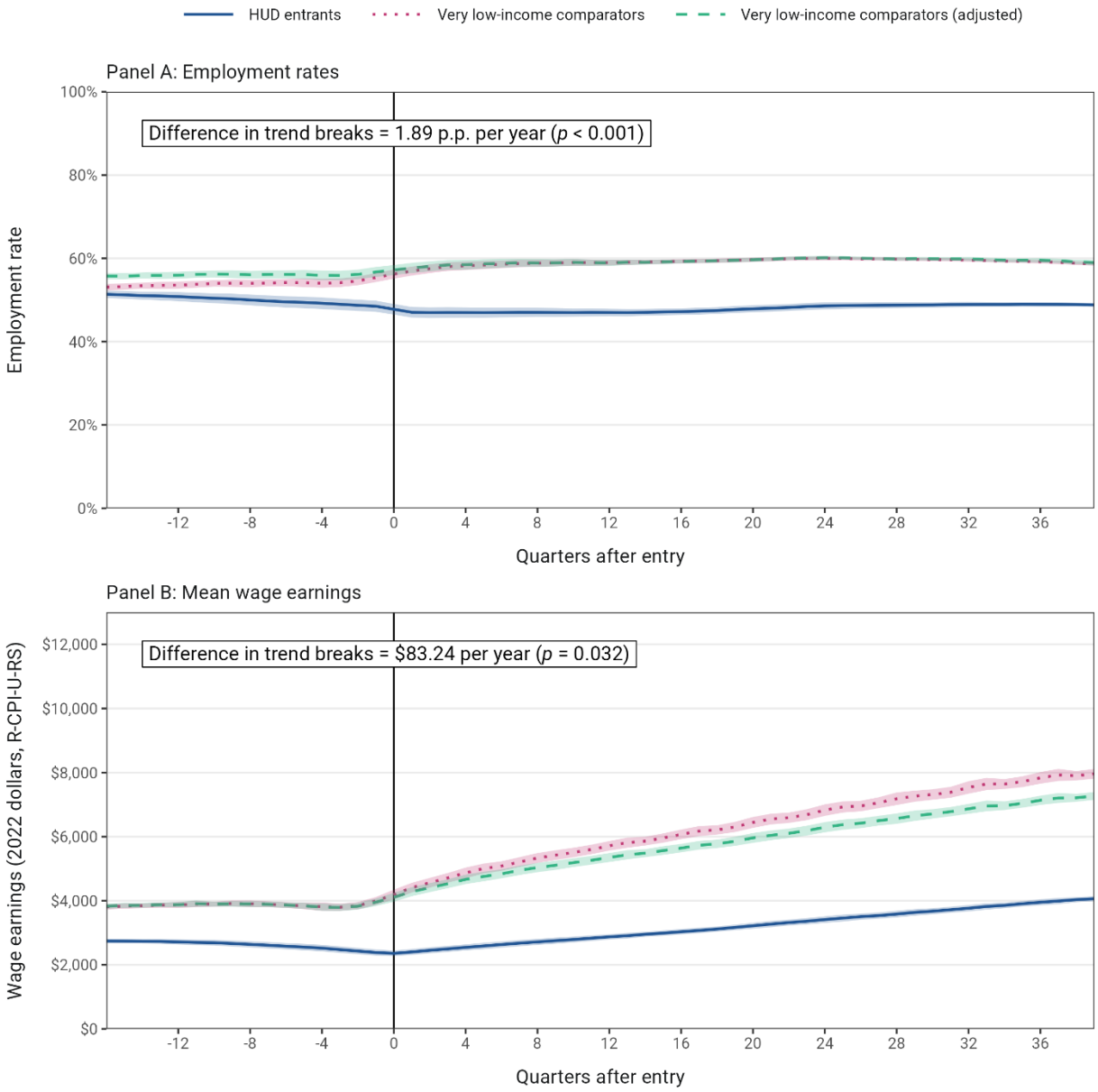
Table E.2: Baseline characteristics of HUD entrants and extremely low-income ACS comparators

	HUD entrants	Extremely low-income ACS comparators		Difference (HUD - ACS)	
	Mean	Mean	Adjusted mean	Unadjusted	Adjusted
Baseline outcomes					
Employment rate	50.06%	47.06%	48.71%	3.00 p.p. *	1.35 p.p. *
Quarterly wage earnings	$2,620	$3,112	$3,051	-$492 *	-$431 *
Disability status					
Does not have a disability	80.71%	85.81%	80.71%	-5.10 p.p. *	0.00 p.p.
Has a disability	19.29%	14.19%	19.29%	5.10 p.p. *	0.00 p.p.
Sex					
Female	65.76%	52.48%	65.76%	13.29 p.p. *	0.00 p.p.
Male	34.24%	47.52%	34.24%	-13.29 p.p. *	0.00 p.p.
Race					
Non-Hispanic white	43.04%	46.10%	43.04%	-3.06 p.p. *	0.00 p.p.
Non-Hispanic Black	34.22%	15.68%	34.22%	18.54 p.p. *	0.00 p.p.
Hispanic of any race	16.10%	27.83%	16.10%	-11.73 p.p. *	0.00 p.p.
Non-Hispanic Asian	2.49%	7.21%	2.49%	-4.72 p.p. *	0.00 p.p.
Non-Hispanic two or more races	2.28%	1.69%	2.28%	0.58 p.p. *	0.00 p.p.
Non-Hispanic AIAN	1.29%	0.93%	1.29%	0.36 p.p. *	0.00 p.p.
Non-Hispanic other race	0.57%	0.56%	0.57%	0.02 p.p.	0.00 p.p.
Age					
Age at entry or interview (years)	34.75	34.45	34.75	0.30 *	0.00
Census region					
South	37.30%	33.55%	37.30%	3.75 p.p. *	0.00 p.p.
Midwest	24.82%	20.30%	24.82%	4.52 p.p. *	0.00 p.p.
Northeast	19.03%	18.02%	19.03%	1.01 p.p. *	0.00 p.p.
West	18.85%	28.13%	18.85%	-9.28 p.p. *	0.00 p.p.
Household characteristics					
Household income	$19,160	$14,330	$13,210	$4,827 *	$5,952 *
Number of adults	1.45	1.80	1.45	-0.35 *	0.00
Number of young children (0-4)	0.53	0.38	0.53	0.15 *	-0.00
Number of children (5-12)	0.58	0.49	0.58	0.09 *	-0.00
Number of older children (13-17)	0.22	0.23	0.22	-0.01 *	0.00
HUD program					
Housing Choice Vouchers	46.81%				
Public housing	27.22%				
Project-based Section 8	22.72%				
Other multifamily	3.26%				
Sample sizes					
Observations	2,086,000	458,000	458,000	2,545,000	2,545,000
Unique PIKs	2,086,000	455,000	455,000	2,541,000	2,541,000

Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Extremely low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's extremely low-income eligibility limit (set at 30% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Baseline outcomes are measured during the 16 quarters leading up to the quarter of entry or interview. All dollar amounts are adjusted to 2022 levels using the R-CPI-U-RS. Adjusted statistics are obtained by reweighting comparators to match the mean non-outcome characteristics of HUD entrants within each quarterly entry cohort. Asterisks (*) indicate that an estimated difference is statistically significant at the 5-percent level, based on a statistical test that accounts for heteroskedasticity and clustering at the PIK level. AIAN = American Indian/Alaska Native.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

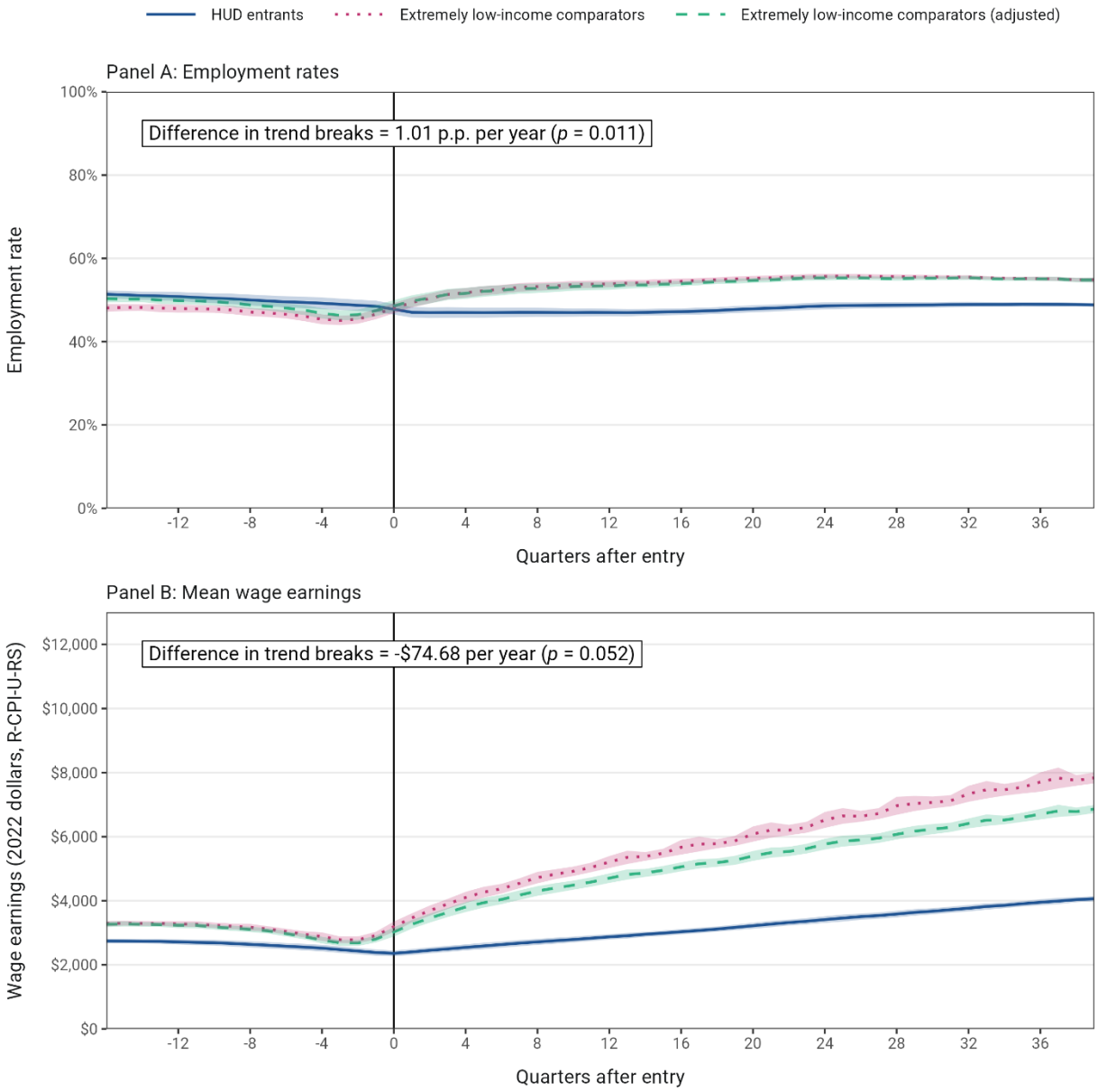
Figure E.1: Labor market trajectories of HUD entrants and very low-income ACS comparators



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Very low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's very low-income eligibility limit (set at 50% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).

Figure E.2: Labor market trajectories of HUD entrants and extremely low-income ACS comparators



Notes: HUD entrants are working-age adults who entered a HUD program for the first time between 2005 Q1 and 2014 Q1. Extremely low-income ACS comparators are working-age adults who have no record of participating in a HUD program as of 2022, but otherwise met HUD's extremely low-income eligibility limit (set at 30% of Area Median Income with adjustments for family size) when they responded to the ACS between 2005 Q1 and 2014 Q1. Adjusted trajectories are obtained by reweighting comparators to match the mean non-outcome characteristics of entrants within each quarterly entry cohort. Vertical black lines mark the quarter entrants started participating in HUD programs. Blue, magenta, and teal shaded areas outline 95-percent confidence intervals that are robust to heteroskedasticity and serial correlation within entry cohorts.

Data sources: HUD PIC/TRACS, ACS, LEHD, MAF-X, MAF-ARF, Census Numident, and the Decennial Census. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505232: CBDRB-FY26-028).